

BZAN 535: Statistical Methods

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Preface

BZAN 535 Study Book

Chapter Overview

Chapter 1 — Role of Statistics and Key Concepts

Introduces the role of statistics in turning data into informed decisions. Covers data types and measurement scales, populations and samples, parameters and statistics, descriptive versus inferential statistics, and the distinction between observational studies and randomized experiments. Particular attention is given to the different roles of random sampling and random assignment in supporting generalization and causal inference.

Chapter 2 — Descriptive Analysis

Develops the tools used to summarize and understand observed data. Covers measures of center, spread, and distribution shape; frequency tables; outliers; and common visualizations including histograms, boxplots, scatterplots, and bar charts. The chapter also introduces the `ggplot2` framework for constructing statistical visualizations in R.

Chapter 3 — Descriptive Analytics for Time Series Data

Extends descriptive analysis to data observed over time. Introduces date and time manipulation with `lubridate`, time-based filtering and grouping, and visualization of temporal data. The chapter examines important time-series patterns including trend, seasonality, short-term variation, and level changes, along with techniques such as moving averages for revealing underlying patterns.

Chapter 4 — Probability Rules and Distributions

Moves from describing observed data to reasoning about uncertain outcomes. Introduces probability terminology, probability rules, conditional probability, independence, and the distinction between discrete and continuous random variables. The chapter then develops four important probability distributions—Binomial, Normal, Poisson, and Exponential—and explains how to recognize, interpret, and work with each distribution in R.

1 Role of Statistics and Key Concepts

1.1 Overview

- A decision before definitions
- Role of statistics in business
- Data types and measurement scales
- Populations vs. Samples
- Parameters vs. Statistics
- Study Designs: Observational vs. Experimental

Statistics: Business Question → Data → Analysis → Decision

The goal is to make that pipeline rigorous, not a simple “gut call.” There should be some method behind the process.

- Precisely describe what the data shows. *What was the average order value?*
- Estimate what’s outside the data with honest uncertainty. *What does our sample tell us about customers in general?*
- Compare groups within data and decide if differences are real or noise. *Do customers exposed to different policies actually behave differently?*
- Model relationships and determine if they’re causal. *What variables are related to spending and do they actually cause some sort of change?*

1.2 Types of Data

Categorical: qualitative values relating to characteristics.

- **Nominal:** categories with **no meaningful order**. There’s no mathematical sense in which is greater or less than another. *e.g. red, blue, green*
- **Ordinal:** categories **with an order** to them. Certain categories correspond to higher or lower rankings, places, etc. *e.g. one star, two star, three star*

Numerical

- **Interval:** numerical values with meaningful differences but no true zero. A value of zero doesn't necessarily mean "none", rather it's just a 'label' similar to any other number. *e.g. $0^{\circ}F$, $53^{\circ}F$*
- **Ratio:** numerical values with a true equal spacing and a true zero. In this case, a value of zero literally means "none" or "nothing there". *e.g. \$0, \$5* Because zero has an actual meaning, comparisons (or ratios) like "twice as much" are relevant.

1.3 Population vs. Sample

- Population: entire group of interest. This is the true group in question.
- Sample: the limited subset being observed.
- Parameter: a description of the population. Because it involves the entire group of interest, it's unknown (it's impossible to actually include everyone).
- Statistic: a computed description of the sample. Because the sample is only a part of the group, meaning it's an inference that it represents the true group, it's a predicted value.

	Population	Sample
Described by	Parameter	Statistic
Center	μ	$\bar{x}$
Proportion	p	$\hat{p}$
Spread	σ	s

Say in reality there's 50,000 customers and you survey 40. The average satisfaction score of the survey is 4.2. What you really want to know is $\mu = ?$ – the average satisfaction of all 50,000 customers.

Because only 40 were surveyed, you're using statistic $\rightarrow$ estimate parameter to estimate μ . That's the foundation of **statistical inference**.

Taste a spoon of soup (sample) to judge the pot (population), but only if you stirred it first (representative sample).

1.4 Two Broad Branches of Statistics

Descriptive vs. Inferential – that distinction becomes important.

Descriptive: describe the data actually observed. Things like mean, median, standard deviation, histogram, boxplot.

The customers in our sample spent an average of \$75.

That's descriptive

Inferential: use your sample to say something about a larger population. Things like confidence intervals, hypothesis tests, regression, experiments.

Based on the sample, we estimate RetailCo customers spend...

Assuming reality: inferential.

Goal	Tools	Output Examples
Describe Infer / Predict	Tables, histograms, boxplots Confidence intervals, hypothesis tests, regression, machine learning	Center, spread, shape Effects, forecasts

1.5 Statistical Tools

Question	Data	Tool
What's typical/variable?	Quantitative	Mean, SD, IQR, histogram, boxplot
How many are in each group?	Categorical	Frequency table, bar chart
Do groups differ?	Categorical + quantitative	Side-by-side boxplots, summaries
Are two quantities associated?	Two quantitative variables	Scatterplot, correlation

So you don't just randomly choose the statistical technique, the business question and variable types determine the appropriate tool.

1.6 Study Designs: Observational vs. Experimental

Suppose a company claims customers who use our app spend 40% more. Does that mean our app causes customers to spend more? Not necessarily. Many people who download the app could already be the company's most loyal customers. This is the problem of **confounding**.

Observational: observe what people already chose/did. Here, you can establish an association, but causality isn't established merely by observing a difference.

Experiment (A/B): random assignment of subjects. This randomization helps balance other characteristics (lurking variables) between groups. Therefore, differences in outcomes can support a causal interpretation, assuming everything else in the experiment is valid.

Randomized experiments can support cause-and-effect inference, whereas observational studies do not by themselves; random sampling is separately what supports inference to a population.

Random Sampling and random assignment are **not** the same thing.

- Random sampling answers “Can I generalize this result to the population?”.
- Random Assignment answers “Can I make a causal claim?”

Concept	What Is Randomized?	When?	Purpose
Random Sampling	Which people from the population are selected for the sample	When selecting a sample	Makes the sample more representative of the population and supports generalizing to the population
Random Assignment	Which treatment/group participants receive	When conducting an experiment	Makes treatment groups more comparable and supports stronger causal conclusions

2 Descriptive Analysis

2.1 Overview

Raw data can be messy and overwhelming, making it difficult to interpret and work with. Descriptive analysis helps solve this problem by summarizing and exploring data in a way that turns raw numbers into patterns and insights that can be understood accurately and easily.

There are three key questions to consider:

- What's typical?
- How spread out is it?
- What shape does it have?

How do we answer these questions? By examining the center, spread, and shape of the data – each of which describes a different aspect of the observations and together, helps us draw accurate conclusions about what the data is telling us.

Descriptive Analysis is organized around several key ideas:

- Center — What is typical?
- Spread — How much does the data vary?
- Shape — What does the distribution look like?
- Groups — How do different categories or populations compare?
- Visualization — How can we make patterns easier to see?

The first three, center, spread, and shape, describe the distribution itself. Groups lets you compare distributions, while visualization is how you explore and communicate those characteristics. And all are equally important.

A Quick Example: Suppose two products both sell an average of 100 units per week:

- Product A: 98, 101, 102, 99, 100 → extremely consistent
- Product B: 20, 180, 40, 160, 100 → highly variable

Both products have the same average, but their weekly sales behave very differently. Knowing the average alone does not necessarily tell us what sales actually look like.

They have the same center, but very different spread.

Together, these tools help answer the key questions needed to understand a dataset before moving on to deeper analysis.

2.2 Center

What's a typical value?

Center is the measure that represents the typical or central value of a dataset. It describes the location around which the data tends to lie.

There are two main measures of center: mean and median.

2.2.1 Mean

The **mean** is the arithmetic average of a dataset. It is calculated by adding all observations together and dividing by the number of observations.

$$\bar{x} = \frac{\sum x_i}{n}$$

Pros:

- Uses every observation in dataset.
- Easy to calculate and interpret.

Cons:

- Sensitive to outliers.
- Can be misleading for highly skewed distributions.

2.2.2 Median

Median is the middle value of a dataset after the observations have been ordered. It is more robust to outliers than the mean. For a odd number of observations, sort the data and take the middle value. For an even number of observations (i.e. two middle values) take the two middle values and average them.

Pros:

- Robust to outliers and extreme values.
- Often better represents the center of skewed data.

Cons:

- Does not incorporate the magnitude of every observation the same way the mean does.
- Can ignore meaningful differences in the values away from the middle.

2.3 Spread

How much do values vary?

Spread describes how much the observations in a dataset vary or differ from one another. In other words, it tells us how dispersed the data is around its center.

2.3.1 Range

The range measures the distance between the smallest and largest observations.

$$\text{range} = \text{maximum} - \text{minimum}$$

Pros:

- Extremely simple to calculate or understand.

Cons:

- Only considers two observations – the largest and smallest – so it is highly sensitive to outliers.

2.3.2 IQR

The **IQR** measures the spread of the middle 50% of the observations. In other words, how much do the middle 50% of values span? Because it ignores the extreme ends, it's resilient to outliers.

$$\text{IQR} = Q_3 - Q_1$$

Q_1 is the 25th percentile. Q_3 is the 75th percentile.

Conceptually, IQR looks at:

$$\underbrace{\text{lowest 25\%}} \quad | \quad \underbrace{Q_1 \leftrightarrow Q_3}_{\text{middle 50\%}} \quad | \quad \underbrace{\text{highest 25\%}}$$

Pros:

- Robust to outliers.

Cons:

- Does not incorporate all observations.

2.3.3 Variance

Variance measures how spread out observations are around the mean using their squared deviations from the mean. The deviations are squared so that negative and positive deviations do not cancel each other out.

To calculate:

1. Find the mean.
2. Find how far each observation is from the mean.
3. Square each difference.
4. Add the squared differences.
5. Divide by $n-1$.

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Pros:

- Uses every observation in the dataset.
- Measures spread specifically around the mean, so it pairs naturally with the mean.

Cons: - Expressed in squared units, meaning it is less intuitive.

2.3.4 Standard Deviation

Standard Deviation is the square root of the variance. It allows for an interpretable scale of how much values differ from mean by putting variance back into original units.

$$s = \sqrt{s^2}$$

Pros:

- Easy to interpret.

Cons:

- Its magnitude depends on the scale of the data and can be misleading when comparing two variables of different scales.
- Sensitive to outliers.

2.3.5 Coefficient of Variation

The **Coefficient of Variation** measures the standard deviation relative to the mean. In other words, it tells you how large the standard deviation is compared to the mean (how big is distance based on mean).

$$CV = \frac{SD}{\bar{x}}$$

Pros:

- Can accurately compare spread of variables with different scales.

Cons:

- Not useful when mean is zero, because you cannot divide by zero.
- Can be sensitive to outliers.

2.4 Shape

What does the distribution look like?

Shape describes the overall pattern or form of a distribution. It helps us understand how observations are distributed around the center, including whether the data is symmetric, skewed, or contains outliers.

2.4.1 Symmetric

A **symmetric** distribution has approximately the same shape on both sides of its center. Neither side has a substantially longer tail than the other.

In a symmetric distribution, the mean and median tend to be relatively close.

2.4.2 Right Skewed

A **right-skewed** distribution has a long tail extending to the right, toward larger values.

A relatively small number of large observations stretch the distribution to the right and can pull the mean upward.

Typically: right skewed $\rightarrow$ mean $>$ median

2.4.3 Left Skewed

A **left-skewed** distribution has a long tail extending to the left, toward smaller values.

A relatively small number of small observations stretch the distribution to the left and can pull the mean downward.

Left tail → mean pulled left → Mean < Median Right tail → mean pulled right → Mean > Median

2.4.4 Outliers

An **outlier** is an observation that is unusually far from the rest of the data.

The **IQR rule** can be used to identify potential outliers.

lower bound = $Q_1 - 1.5(IQR)$

upper bound = $Q_3 + 1.5(IQR)$

If the observation is outside of these bounds it can be flagged as a potential outlier.

An outlier should not automatically be deleted. It is important to investigate its cause first. For example, an extreme observation could represent a data-entry error, a legitimate unusual observation, or a measurement/unit issue.

2.5 Frequency Tables

A **frequency table** summarizes a **categorical** variable by showing the count and/or proportion of observations in each category. It helps identify common and rare categories and understand the overall category mix.

Department	Count	Percent
Grocery	39,021	42.3%
Drug GM	31,529	34.1%
Produce	3,118	3.4%

2.6 Visualizations

Plot	Data	Main Question
Histogram	1 quantitative variable	What does the distribution look like?
Boxplot	Quantitative, often by category	How does the distribution/spread compare?
Scatterplot	2 quantitative variables	Are the variables related?
Bar chart	Categorical	How do category counts/proportions compare?
Pie chart	Categorical proportions	How is the whole divided among categories?

2.6.1 Histograms

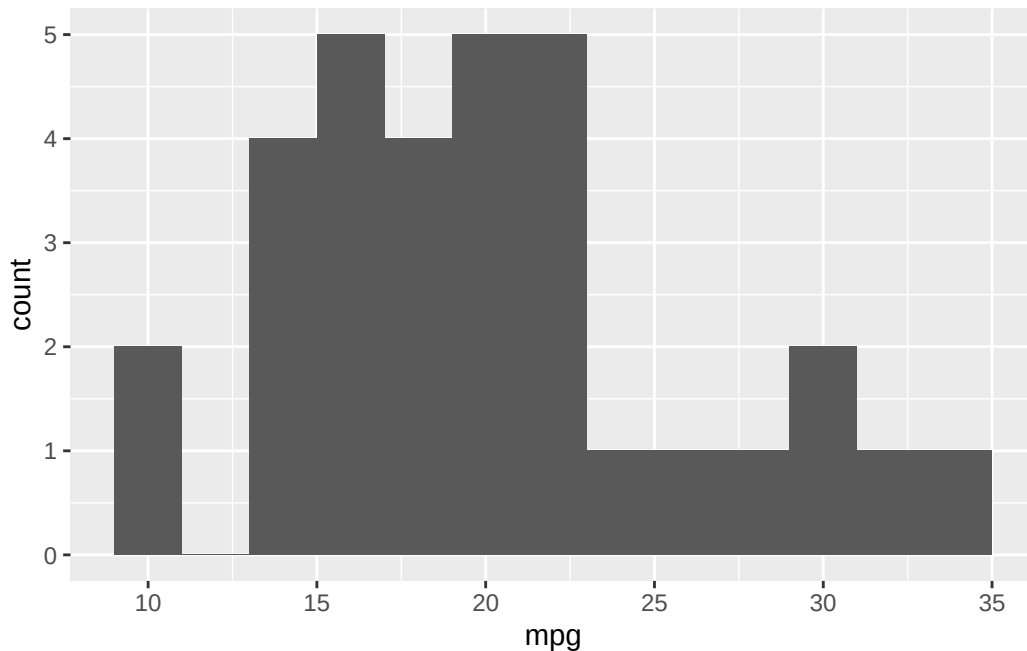
A **histogram** shows the **distribution** of a **single quantitative variable** by grouping numerical values into intervals called bins and showing how many observations fall into each bin.

Useful for identifying:

- Shape of a distribution
- Skewness
- Modes or peaks
- Gaps

The bin width determines how wide each interval is and therefore how much detail the histogram shows

```
ggplot(mtcars, aes(x = mpg)) +  
  geom_histogram(binwidth = 2)
```



Use when: You want to see what the distribution of one quantitative variable looks like.

2.6.2 Boxplots

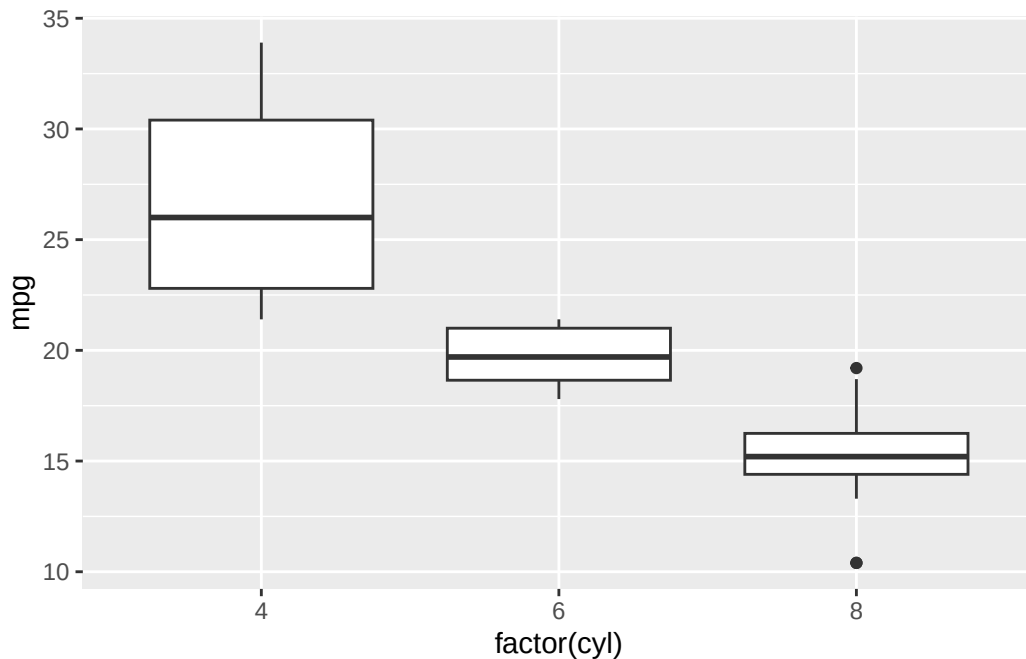
A **boxplot** provides a compact summary of the **distribution** of a **quantitative** variable using the five-number **summary**: Min, Q_1 , Median, Q_3 , Max

It is useful for examining:

- Center
- Spread
- Skewness
- Potential outliers
- Differences between groups

Boxplots are especially useful when you want to **compare distributions** across categories.

```
ggplot(mtcars, aes(x = factor(cyl), y = mpg)) +  
  geom_boxplot()
```



Use when: You want to examine or compare the distributions of quantitative data, particularly across groups.

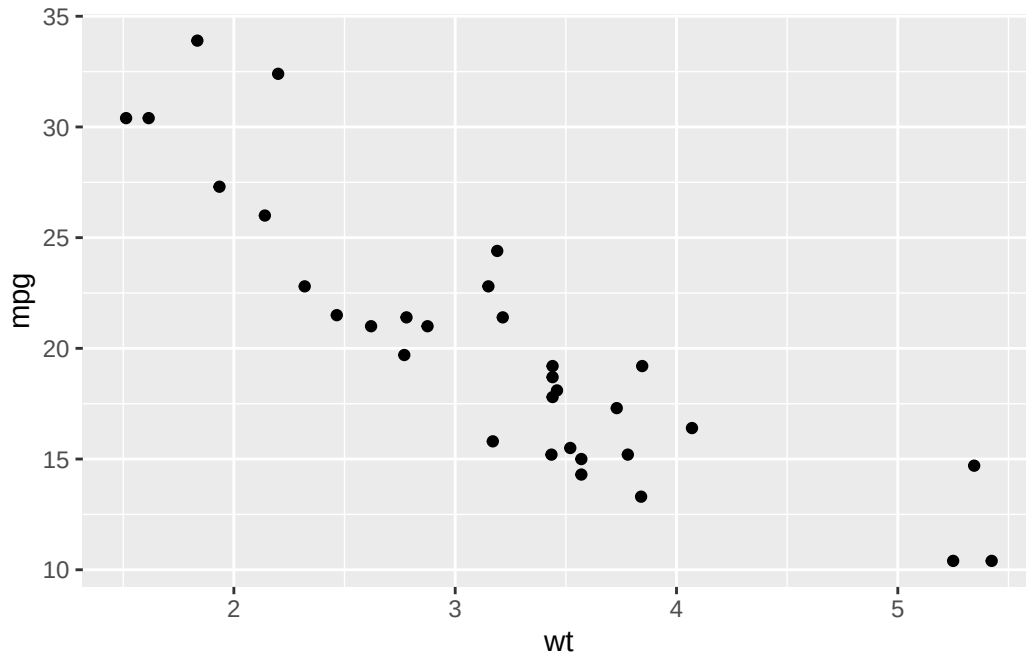
2.6.3 Scatterplots

A **scatterplot** shows the **relationship** between **two quantitative variables**. Each point represents one observation.

Useful for identifying:

- Direction — positive or negative relationship
- Strength — how closely the variables are related
- Form — linear, nonlinear, etc.
- Outliers

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point()
```



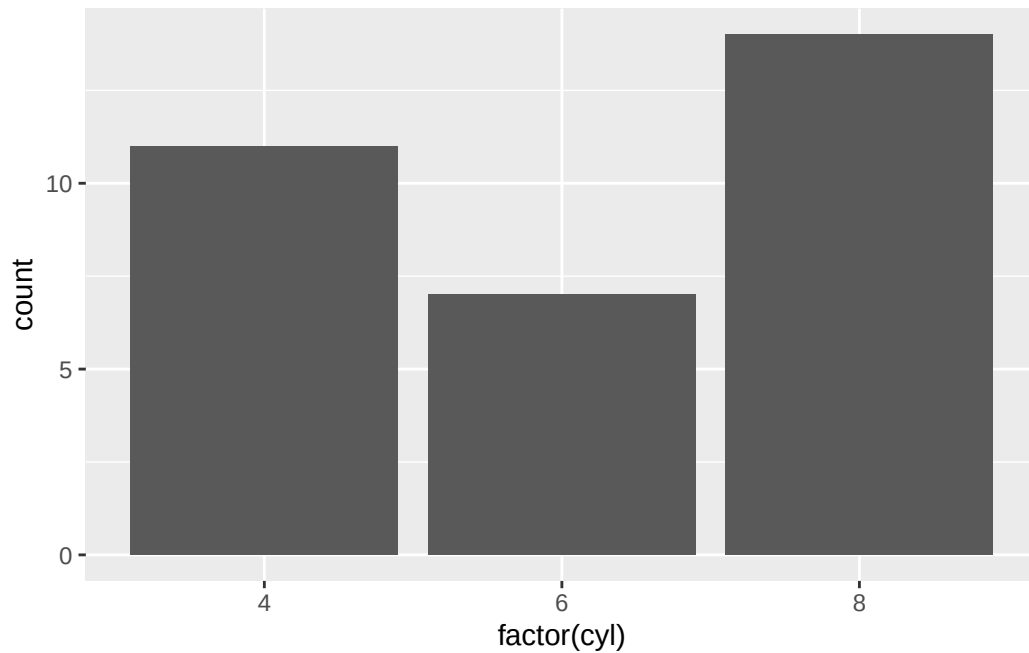
Use when: You want to know whether two quantitative variables are related.

2.6.4 Bar Charts

A **bar chart** displays the counts or proportions of categories for a categorical variable. It is essentially the graphical version of a **frequency table**.

The categories are represented by bars, and the height of each bar represents its count.

```
ggplot(mtcars, aes(x = factor(cyl))) +  
  geom_bar()
```



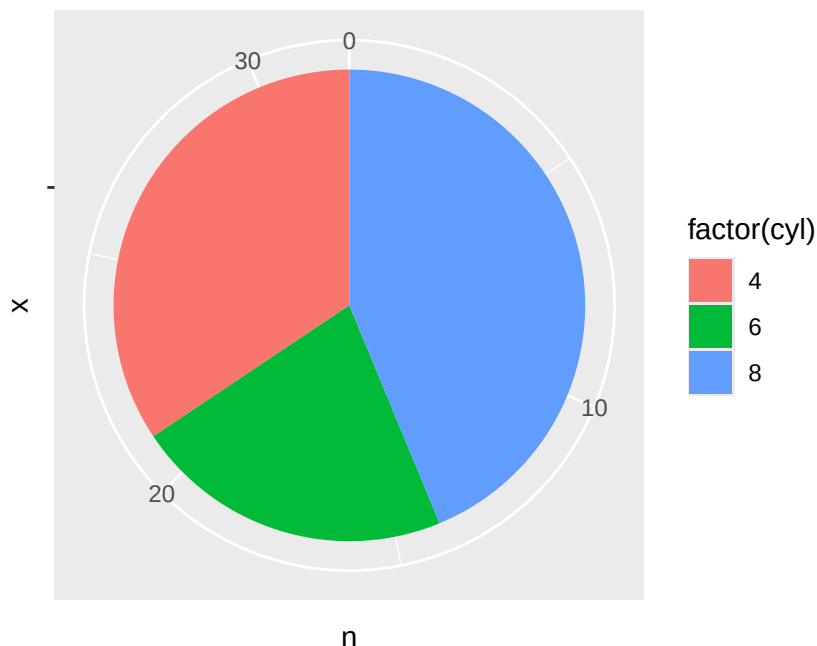
Use when: You want to compare counts or proportions across categories.

2.6.5 Pie Charts

A **pie chart** displays **proportions of a whole** as slices of a circle. Larger slices represent larger proportions.

Pie charts work best when there are only a **small number of categories**. Similar-sized slices can be difficult to compare visually, so a bar chart is often easier to interpret.

```
mtcars |>
  count(cyl) |>
  ggplot(aes(x = "", y = n, fill = factor(cyl))) +
  geom_col(width = 1) +
  coord_polar(theta = "y")
```



Use when: You want to show how a small number of categories make up a whole.

2.7 ggplot2

ggplot2 is a data visualization package in R built by combining data, aesthetics, and layers.

data + aesthetics + layers

Data = what you're plotting **aes()** = which variables control visual properties **geom_*()** = how the observations are drawn **scale_*()** = how values are displayed **facet_*()** = split into groups **labs()** = titles and labels **theme_*()** = appearance

2.7.1 Aesthetics

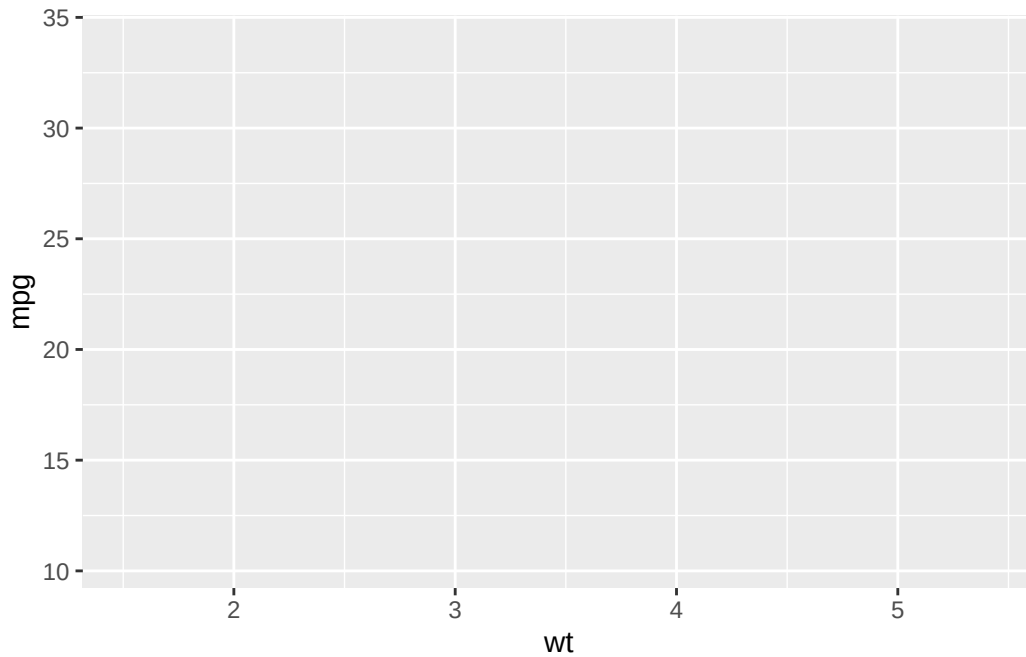
Aesthetics, specified with **aes()**, determine how variables in the data **are mapped to visual properties** of the graph.

Common aesthetics include:

- x — horizontal position
- y — vertical position
- color — color
- size — size

- shape — shape
- alpha — transparency

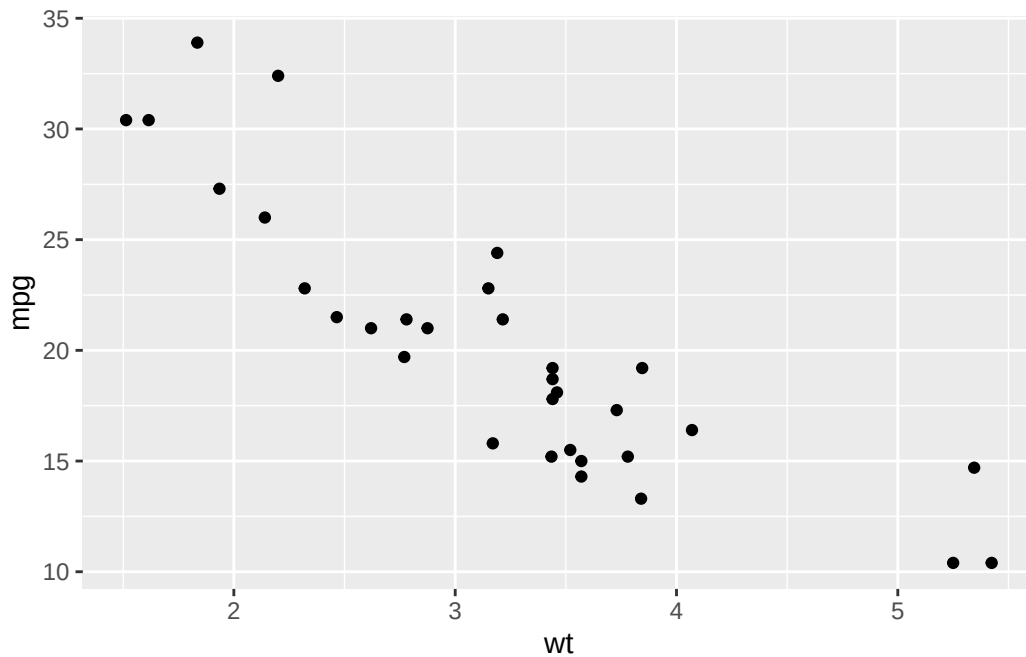
```
ggplot(mtcars, aes(  
  x = wt,  
  y = mpg,  
  color = factor(cyl)  
))
```



2.7.2 Layers

Layers determine how the data is actually displayed. They are usually added with `geom_*()` functions using `+`.

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point()
```



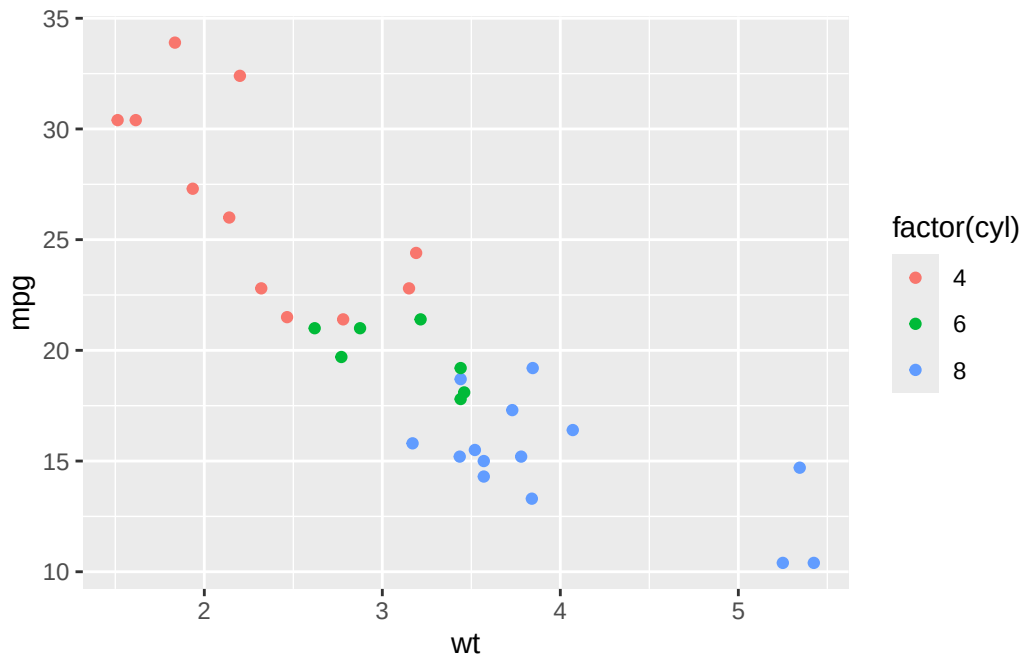
Examples include:

`geom_point()` `geom_histogram()` `geom_boxplot()` `geom_col()`

2.7.3 Mapping vs. Setting

Mapping means a visual property *varies* according to a variable in the data. Put it inside `aes()`

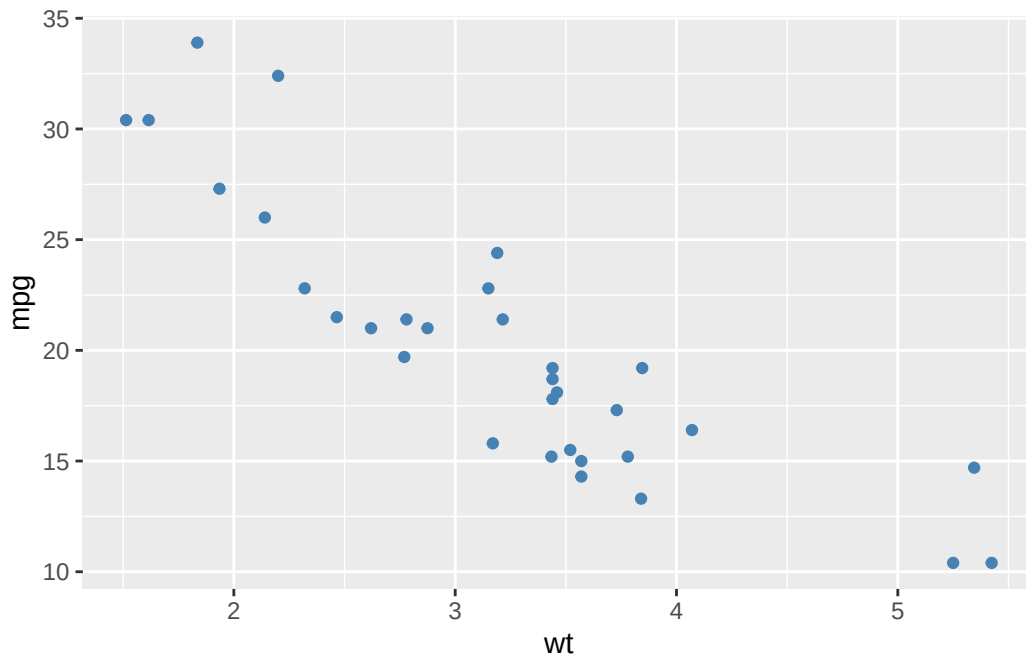
```
ggplot(mtcars, aes(  
  x = wt,  
  y = mpg,  
  color = factor(cyl)  
)) +  
  geom_point()
```



Different household-size categories receive different colors.

Setting means giving a visual property one fixed value. Put it outside `aes()`

```
ggplot(mtcars, aes(  
  x = wt,  
  y = mpg  
) +  
  geom_point(color = "steelblue")
```

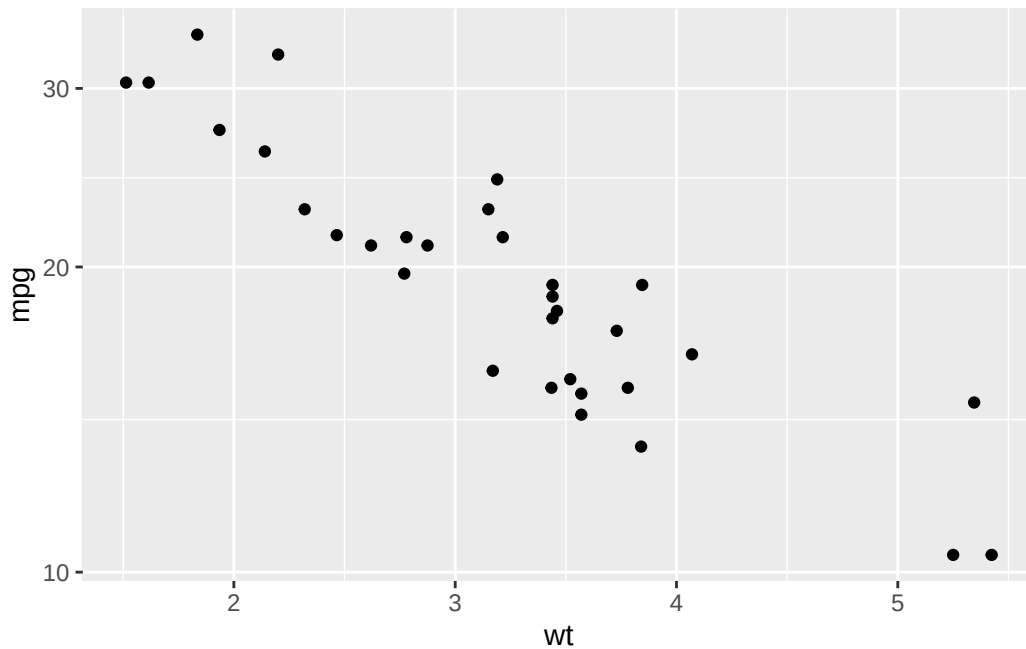


Now every point is steel blue.

2.7.4 Scales

Scales control how mapped data values are displayed, such as transforming or formatting an axis.

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  scale_y_log10()
```

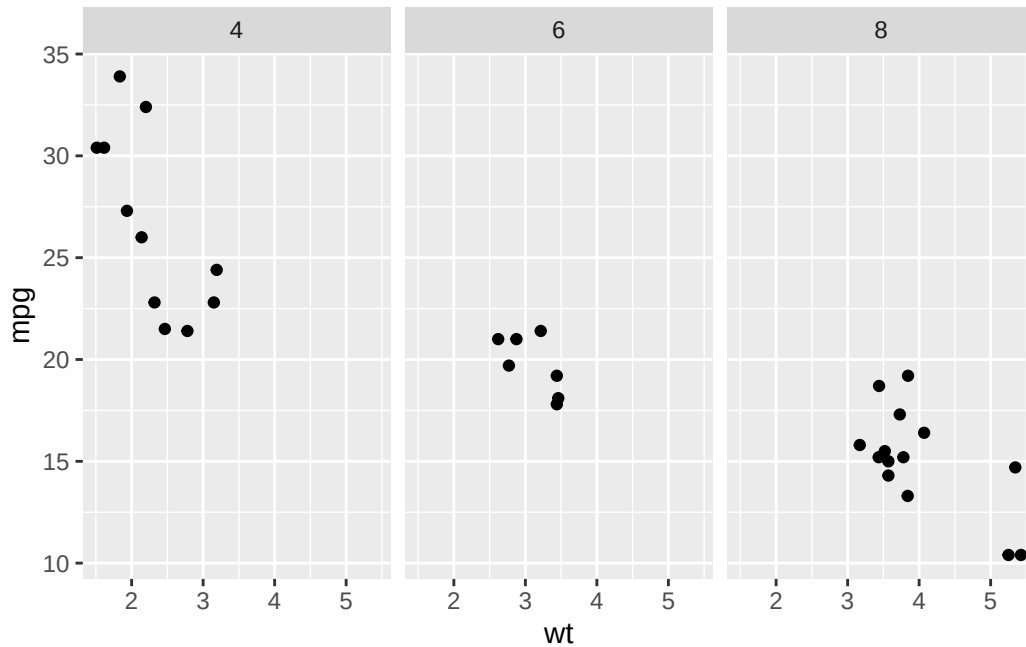


`scale_y_log10()` displays the y-axis on a logarithmic scale.

2.7.5 Facets

Facets split one visualization into separate plots for different groups.

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  facet_wrap(~ cyl)
```

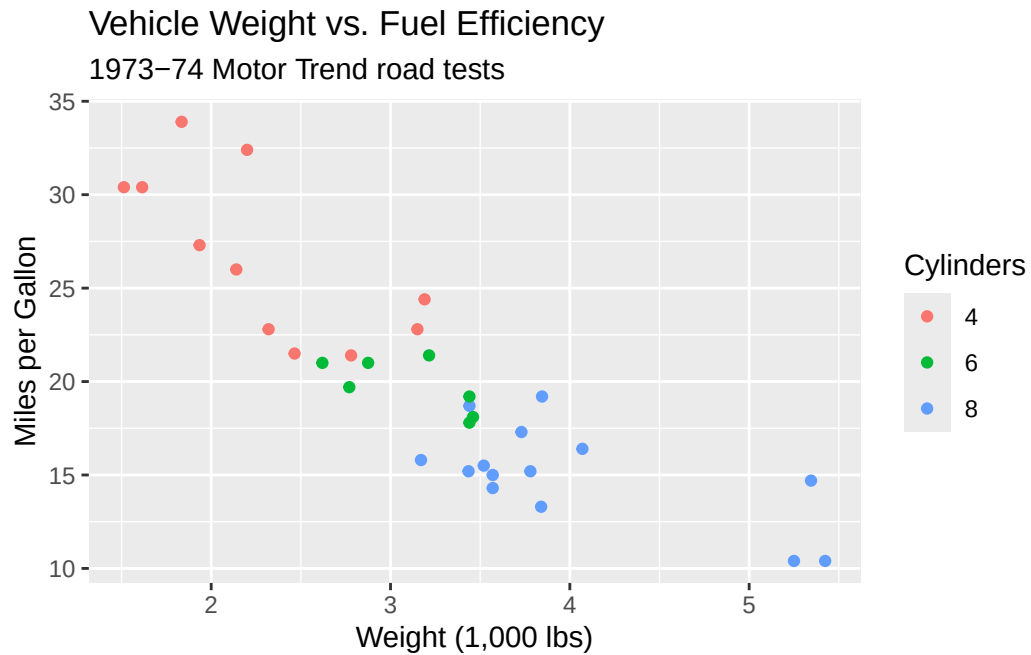


Instead of showing every household-size group on one graph, you get a separate panel for each household size.

2.7.6 Labels

`labs()` controls the descriptive text on the visualization:

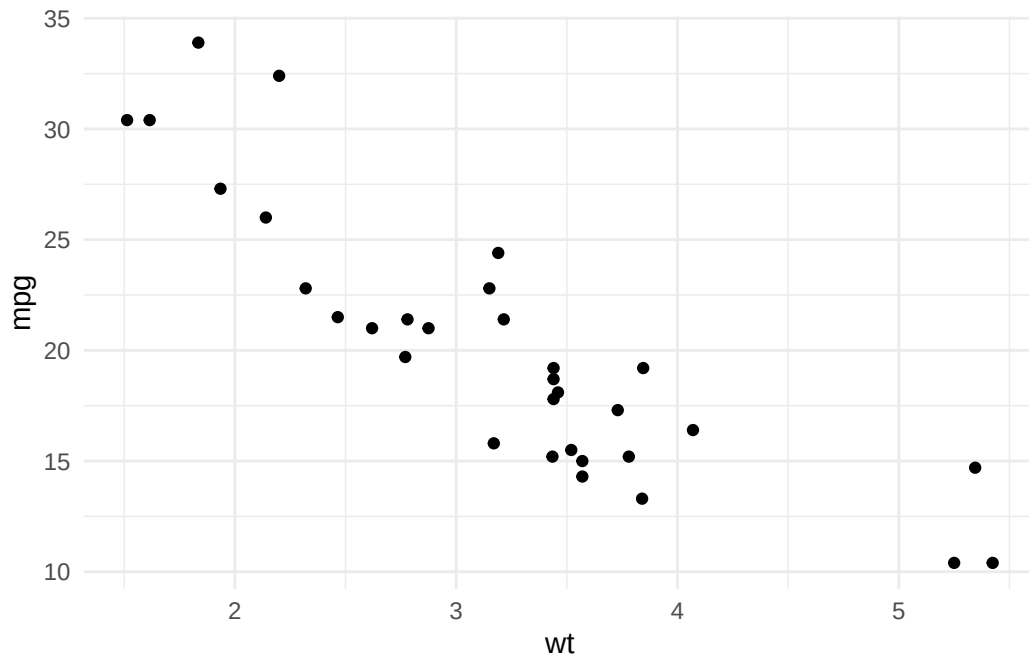
```
ggplot(mtcars, aes(x = wt, y = mpg, color = factor(cyl))) +
  geom_point() +
  labs(
    title = "Vehicle Weight vs. Fuel Efficiency",
    subtitle = "1973-74 Motor Trend road tests",
    x = "Weight (1,000 lbs)",
    y = "Miles per Gallon",
    color = "Cylinders"
  )
```



2.7.7 Themes

Themes control the overall appearance of the plot rather than the underlying data.

```
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point() +  
  theme_minimal()
```



3 Descriptive Analytics for Time Series Data

3.1 Overview

For descriptive analysis, the focus was on questions such as: What is typical? How spread out is the data? What shape does the distribution have? With time series data, an additional question becomes important: When did it happen?

Time series data is a sequence of observations collected or recorded at successive points in time. Because observations are ordered chronologically, time series data provides a dynamic view of how a variable changes over time, rather than a static snapshot.

- Chronological order: Observations occur sequentially over time so ordering of the data matters.
- Time index: Each observation is associated with a specific point in time.
- Dependence over time: Earlier and later observations may be related, meaning observations may not be independent.
- Temporal patterns: The ordering of observations allows patterns like trend, seasonality, and short term variation to be identified.

Suppose a CFO says “Sales dropped 15% from Q3 to Q4.”

That number by itself doesn’t necessarily mean the business suddenly got worse. Maybe sales normally decline from Q3 to Q4 and this is just a regular reoccurring observation.

With time series data, there are three things to be distinguished:

- Trend → Is there a long-term upward/downward movement?
- Seasonality → Does a pattern repeat at regular intervals?
- Noise → Is a movement just short-term random variation?
- Level Change → Did the series actually shift to a different level?

3.2 Working with Lubridate

Why do dates matter in analytics? Dates can be used for several things, for example:

- Time aware filtering (e.g. time based filtering), time aware grouping (group a time period), and trend/seasonality detection.

- Aligning events (comparing before and afters).
- Preparing for forecasting.

Dates need to actually be dates! A date stored as text like:

```
'2025-08-21'
```

is much different from an R `Date` object actually representing August 21, 2025. The main package used when working with dates is `lubridate`.

3.2.1 Parsing Dates

Parsing dates is the process of converting raw string dates to an actual date/time object, which can then be converted into useful time components. Raw string dates may not always be written in a consistent way, therefore different functions are used based on what the text format is.

- `ymd()` when string = YEAR → MONTH → DAY
- `mdy()` when string = MONTH → DAY → YEAR
- `dmy()` when string = DAY → MONTH → YEAR
- `ymd_hms()` when string = YEAR → MONTH → DAY : HH:MM:SS

Example:

```
ymd("2025-08-21")
```

```
[1] "2025-08-21"
```

```
mdy("08/21/2025")
```

```
[1] "2025-08-21"
```

```
dmy("21-Aug-2025")
```

```
[1] "2025-08-21"
```

```
ymd_hms("2025-08-21 14:30:00")
```

```
[1] "2025-08-21 14:30:00 UTC"
```

Once the date is in the correct format, R can extract pieces:

- `year(x)` = extract year
- `month(x)` = extract month
- `quarter(x)` = extract quarter
- `wday(x)` = extract weekday
- `hour(x)` = extract hour
- `isoweek(x)` = extract week number of year

```
x <- ymd_hms("2025-08-21 14:30:00")  
x
```

```
[1] "2025-08-21 14:30:00 UTC"
```

```
year(x)
```

```
[1] 2025
```

```
month(x)
```

```
[1] 8
```

```
quarter(x)
```

```
[1] 3
```

```
wday(x)
```

```
[1] 5
```

```
hour(x)
```

```
[1] 14
```

```
isoweek(x)
```

```
[1] 34
```

Essentially the process is **raw date string** → **actual date/time object** → **useful time components**

Dates can also be created from combining variables that already exist. For example if you have `year = 2025` and `month = 8` you can construct an actual date.

`make_date()` use when year/month columns exist because it avoids rounding problems associated with fractional-year representations

```
# Before: year and month are separate variables
txhousing |>
  select(year, month) |>
  head(2)
```

```
# A tibble: 2 x 2
  year month
<int> <int>
1  2000     1
2  2000     2
```

```
# Create an actual date from year and month
TX <- txhousing |>
  mutate(
    date_actual = make_date(
      year = year,
      month = month,
      day = 1
    )
  )

# After: year and month have been combined into a date
TX |>
  select(year, month, date_actual) |>
  head(2)
```

```
# A tibble: 2 x 3
  year month date_actual
<int> <int> <date>
1  2000     1 2000-01-01
2  2000     2 2000-02-01
```

3.2.2 Calendar Arithmetic

Leap years can screw up calculation of dates. For example `ymd("2024-02-29")` is a leap year. If you subtract five years you'd get 2019-02-29 which doesn't exist.

```
d1 <- ymd("2024-02-29")
```

lubridate has calendar awareness operators:

```
d1 %m-% years(5)
```

```
[1] "2019-02-28"
```

```
ymd("2023-01-31") %m+% months(1)
```

```
[1] "2023-02-28"
```

3.2.3 Time Windows

Once dates are represented properly, normal comparisons work:

```
TX |>
  filter(
    date_actual >= as.Date("2005-01-01"),
    date_actual <= as.Date("2010-12-31")
  )
```

```
# A tibble: 3,312 x 10
```

	city	year	month	sales	volume	median	listings	inventory	date	date_actual
	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<date>
1	Abilene	2005	1	119	1.03e7	74200	570	3.7	2005	2005-01-01
2	Abilene	2005	2	130	1.18e7	74300	581	3.7	2005.	2005-02-01
3	Abilene	2005	3	170	1.45e7	71300	661	4.3	2005.	2005-03-01
4	Abilene	2005	4	171	1.64e7	80300	650	4.2	2005.	2005-04-01
5	Abilene	2005	5	176	1.81e7	87100	660	4.2	2005.	2005-05-01

```

 6 Abilene  2005      6   230  2.41e7  92500      664      4.1 2005. 2005-06-
01
 7 Abilene  2005      7   177  1.85e7  93500      657      4.1 2006. 2005-07-
01
 8 Abilene  2005      8   216  2.41e7  97500      638      3.9 2006. 2005-08-
01
 9 Abilene  2005      9   165  1.71e7  92800      644      3.9 2006. 2005-09-
01
10 Abilene  2005     10   161  1.64e7  92300      601      3.6 2006. 2005-10-
01
# i 3,302 more rows

```

You can also make relative windows.

For example, find the most recent date:

```

max_date <- max(TX$date_actual, na.rm = TRUE)
max_date

```

```
[1] "2015-07-01"
```

Then go five years backward:

```

start_5y <- max_date %m-% years(5)
start_5y

```

```
[1] "2010-07-01"
```

Then:

```

recent <- TX |>
  filter(date_actual > start_5y)

head(recent)

```

```

# A tibble: 6 x 10
  city      year month sales    volume median listings inventory  date date_actual
<chr>   <int> <int> <dbl>    <dbl>   <dbl>    <dbl>    <dbl> <dbl> <date>
1 Abilene  2010     8   144 17504580 122000     936     6.7 2011. 2010-08-
01

```

2	Abilene	2010	9	116	15475763	121300	899	6.5	2011.	2010-09-01
3	Abilene	2010	10	111	14570529	111900	863	6.4	2011.	2010-10-01
4	Abilene	2010	11	115	11422741	87900	851	6.4	2011.	2010-11-01
5	Abilene	2010	12	116	15289470	118300	830	6.3	2011.	2010-12-01
6	Abilene	2011	1	68	8834493	123300	809	6.1	2011	2011-01-01

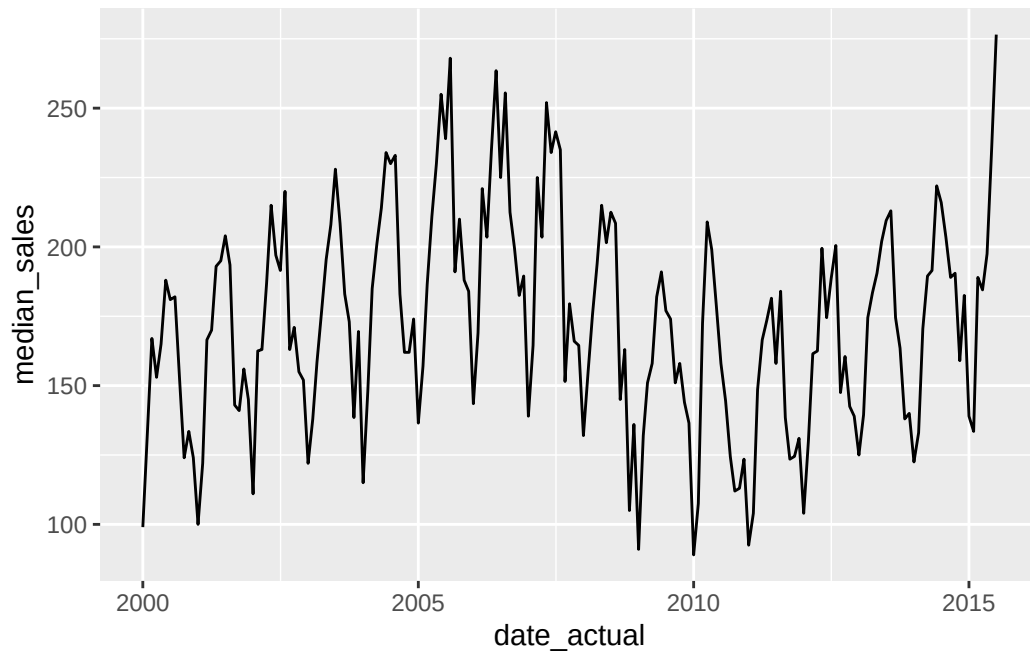
So instead of hard-coding a five-year range, you're saying:

Give me observations from the final five years of the dataset.

For time-series data, the standard visualization is a line chart.

The lecture calculates statewide median sales for each month:

```
TX |>
  group_by(date_actual) |>
  summarise(
    median_sales = median(sales, na.rm = TRUE)
  ) |>
  ggplot(aes(date_actual, median_sales)) +
  geom_line()
```



The conceptual structure is:

- Date → x-axis
- Value → y-axis

Connecting observations makes the evolution of the variable visible.

`scale_x_date()` then controls how frequently dates are displayed so that a long time horizon doesn't produce an unreadable x-axis.

3.2.4 Seasonality

Seasonality is one of the central concepts of time series data.

Seasonality is a pattern that systematically repeats according to the calendar.

It can be viewed by:

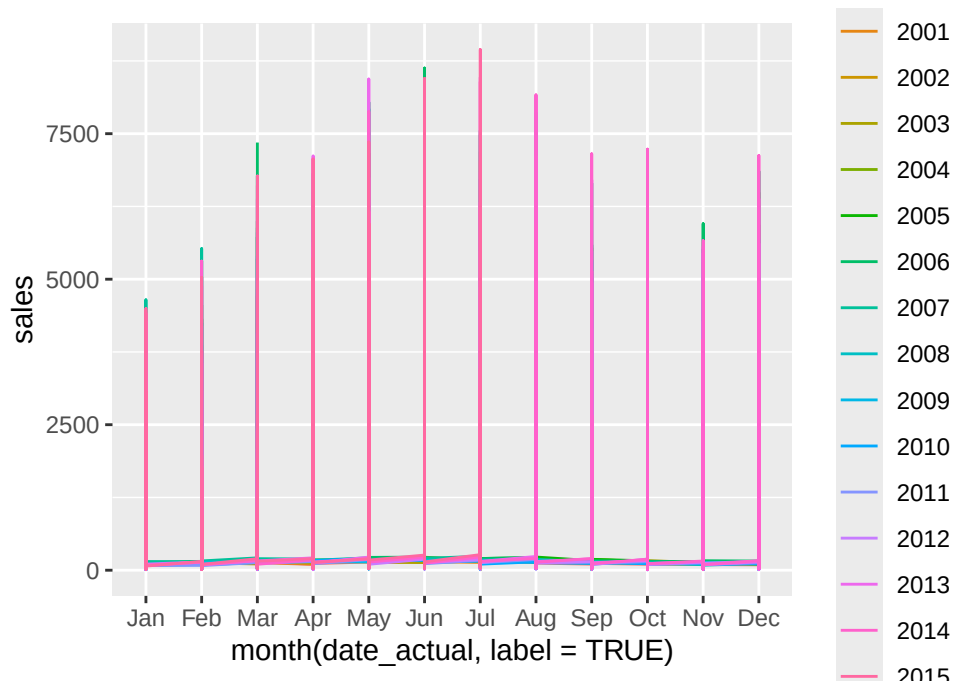
Lines by year:

```
TX |>
  ggplot(aes(
    x = month(date_actual, label = TRUE),
    y = sales,
    group = year,
```

```

    color = as.factor(year)
  )) +
  geom_line()

```



Boxplots by month

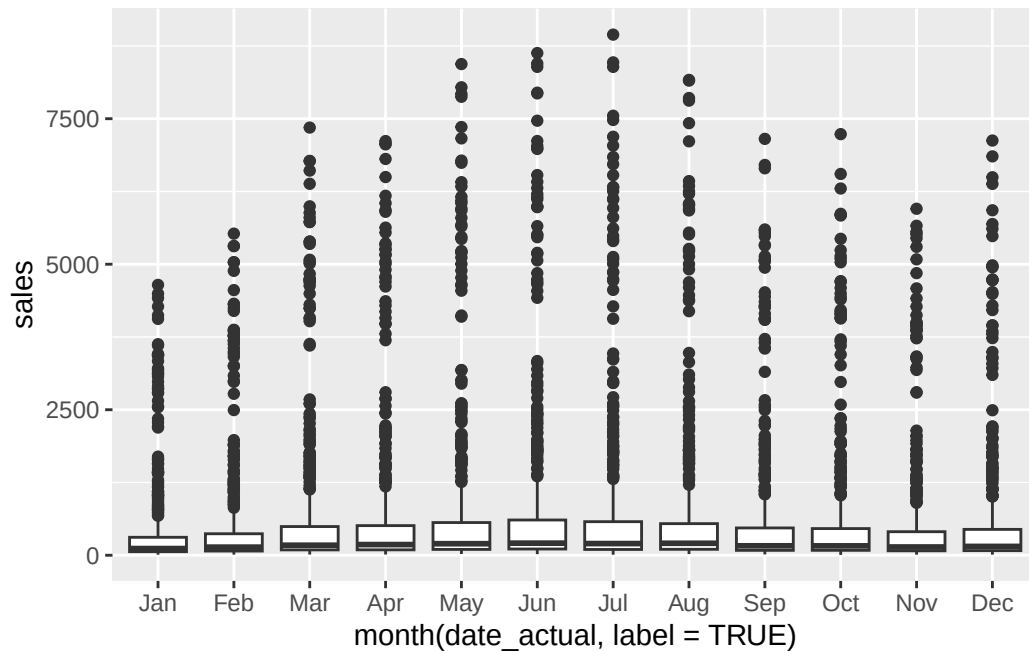
What does a typical January look like across years? What does a typical July look like?

```

TX |>
  ggplot(aes(
    x = month(date_actual, label = TRUE),
    y = sales
  )) +
  geom_boxplot()

```

Warning: Removed 568 rows containing non-finite outside the scale range (`stat\_boxplot()`).



Heatmaps are another view of seasonality

2005 2006 2007 2008 ...

Jan Feb Mar ... Dec

where the fill intensity represents sales.

3.2.5 Moving Averages

Month-to-month fluctuations can make the underlying trend difficult to see. A rolling average smooths those fluctuations.

```
monthly_sales <- TX |>
  group_by(date_actual) |>
  summarise(
    median_sales = median(sales, na.rm = TRUE)
  ) |>
  arrange(date_actual) |>
  mutate(
    moving_average = rollmean(
      median_sales,
```

```

    k = 12,
    fill = NA,
    align = "right"
  )
)
monthly_sales

```

```

# A tibble: 187 x 3
  date_actual median_sales moving_average
  <date>         <dbl>         <dbl>
1 2000-01-01         99          NA
2 2000-02-01        134          NA
3 2000-03-01        167          NA
4 2000-04-01        153          NA
5 2000-05-01        165          NA
6 2000-06-01        188          NA
7 2000-07-01        181          NA
8 2000-08-01        182          NA
9 2000-09-01        152          NA
10 2000-10-01        124          NA
# i 177 more rows

```

$k = 12$ means a 12-observation window. Because the data are monthly, that's a 12-month moving average.

4 Probability Rules and Distributions

4.1 Overview

This section marks the transition from **describing observed data** to **reasoning about uncertain outcomes**. In earlier sections, the focus was on concepts such as center, spread, shape, and trends. Now we extend that framework by asking: “What could happen, and how likely is each outcome?” Answering this requires more than descriptive statistics. It requires looking at **probabilities and probability distributions**.

Things to cover:

- Probability Rules
- Binomial Distribution
- Normal Distribution
- Poisson Distribution
- Exponential Distribution

4.2 Probability

4.2.1 Probability Terminology

A probability is a number from 0 to 1 describing how likely an event is.

Term	Meaning	Example
Experiment	Random process	Send a marketing email
Outcome	One possible result	Customer clicks
Event	One or more outcomes you’re interested in	Customer engages
Sample Space	All possible outcomes	{Click, Open, Ignore, Unsubscribe}

4.2.2 Basic Probability Rules

4.2.2.1 Complement Rule

The complement rule finds the probability that an event does not happen.

$$P(A^c) = 1 - P(A)$$

If event A happens with probability $P(A)$, its complement, written A^c , means “not A.”

For example:

If there's a 20% chance of clicking: $P(\text{No click}) = 1 - .20 = .80$

4.2.2.2 Addition Rule

The addition rule finds the probability that event A OR event B occurs.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

However, that only works if Event A and Event B can occur simultaneously.

If A and B are Mutually Exclusive:

$$P(A \cup B) = P(A) + P(B)$$

For example:

Suppose:

- $P(\text{opens email}) = 0.50$
- $P(\text{clicks link}) = 0.20$
- $P(\text{opens AND clicks}) = 0.15$

Then:

$$P(\text{opens OR clicks}) = 0.50 + 0.20 - 0.15 = 0.55$$

Note: if mutually exclusive: $P(\text{opens OR clicks}) = 0.50 + 0.20 = 0.70$

4.2.2.3 Multiplication

The multiplication rule finds the probability that event A AND event B occur.

If A and B are Dependent Events:

$$P(A \cap B) = P(A) \times P(B | A)$$

If A and B are Independent Events:

$$P(A \cap B) = P(A) \cdot P(B)$$

4.2.3 Conditional Probability

Conditional Probability is the probability an event occurs given another has occurred (i.e. probability of Event A given Event B occurred)

$$P(A | B)$$

4.3 Distributions

4.3.1 Binomial Distribution

A binomial distribution models the number of successes in a fixed number trials, where each trial has two possible outcomes: success or failure. Each trial is independent and the probability of success remains constant.

Discrete

$$X \sim \text{Binomial}(n, p)$$

- $X \rightarrow$ a discrete random variable
- $n \rightarrow$ number of independent trials
- $p \rightarrow$ probability of success

`dbinom(k, n, p)` probability exactly k successes `pbinom(k, n, p)` probability less than or equal to k successes

4.3.2 Normal Distribution

A normal distribution models a continuous random variable with values that are distributed symmetrically around a mean, with values closer to the mean occurring more frequently than values farther from the mean.

mean = median = mode

Continuous

$$X \sim \text{Normal}(\mu, \sigma)$$

- $X \rightarrow$ a continuous random variable
- $\mu \rightarrow$ mean
- $\sigma \rightarrow$ standard deviation

`dnorm(x, mu, sigma)` height of the curve at 100, not probability (continuous variable)
`pnorm(x, mu, sigma)` area to the left of 100, which is a probability

Z-score How many standard deviations away from mean.

$$z = \frac{x - \mu}{\sigma}$$

Empirical Rule

- 68% are within one standard deviation away from mean ± 1
- 95% are within two standard deviations away from mean ± 2
- 99.7% are within two standard deviations away from mean ± 3

Three ways to assess whether data are approximately normally distributed

- Histogram: Shows the shape of the data (roughly symmetric, bell-shaped)
- Q-Q plot: Compares observed data to what we'd expect from a normal distribution (points roughly follow a straight line)
- Shapiro-Wilk test: Formal hypothesis test for normality ($p > 0.05$: do not reject normality)

4.3.3 Poisson Distribution

A Poisson distribution models the number of times an event occurs within a fixed interval of time.

expected number of events (λ) = mean = variance

Discrete

$$X \sim \text{Poisson}(\lambda)$$

- $X \rightarrow$ number of events in the interval
- $\lambda \rightarrow$ average number of events per interval

Function

Calculates

Meaning

- `dpois(k, lambda)` exactly k events
- `ppois(k, lambda)` k or fewer events

Chi-Square Goodness-of-Fit Test A chi-square goodness-of-fit test compares the **observed event counts** to the **expected counts under a Poisson distribution**.

- H_0 : The data are consistent with a Poisson distribution.
- H_A : The data are not consistent with a Poisson distribution.
- $p > 0.05$: Fail to reject $H_0 \rightarrow$ data are reasonably consistent with Poisson.
- $p < 0.05$: Reject $H_0 \rightarrow$ evidence that the data do not fit a Poisson distribution.

Overdispersion Check For a Poisson distribution, the mean and variance should be approximately equal.

$$\frac{\text{Variance}}{\text{Mean}} \approx 1$$

4.3.4 Exponential Distribution

An exponential distribution models the **time between events** in a Poisson process. While the Poisson distribution models **how many events occur in an interval**, the exponential distribution models **how long until the next event occurs**.

Continuous

$$T \sim \text{Exponential}(\lambda)$$

- $T \rightarrow$ waiting time until the next event
- $\lambda \rightarrow$ average rate at which events occur
- `dexp(t, rate = lambda)` density at t
- `pexp(t, rate = lambda)` probability the event occurs within time t
- `qexp(p, rate = lambda)` waiting time associated with cumulative probability p

- `rexp(n, rate = lambda)` generate exponential waiting times

Exponential is memoryless: meaning independent events don't remember what has already happened. What has occurred doesn't impact what is to come.

5 One-Parameter Statistical Inference

5.1 Overview

So far, the focus has been on describing observed data and modeling uncertainty. Descriptive statistics summarize the data we actually observe. Probability distributions describe how uncertain outcomes behave. **Statistical inference** takes the next step: **what can a sample tell us about a population we cannot fully observe?**

Suppose a company reports:

Average household spending last month was \$1,793.

That number may sound precise, but suppose it came from a sample of 2,500 households rather than every household. The sample mean $\bar{x} = 1793$ is known, but the population mean μ is unknown.

The goal of statistical inference is to use the observed **sample statistic** to learn about the unknown **population parameter**, while explicitly accounting for sampling uncertainty.

The general process is:

Population $\rightarrow$ Sample $\rightarrow$ Statistic $\rightarrow$ Inference about Parameter

There are two major tools for doing this:

Confidence intervals Estimate a population parameter

Hypothesis tests Evaluate a claim about a population parameter

Before either one works, however, we need to understand **sampling variability**.

5.2 Sampling

A **sample** is a subset of a population that you actually observe or collect data from. The goal is usually to use information from the sample to learn about the population. This connects to statistics vs. parameters (see Section 1).

Bias The estimate is consistently off from the true value (population) in a particular way, likely caused by poor design.

Variance Sample-to-sample variation that occurs naturally, even with good design. Decreases as n , the number of observations, decreases.

Common sources of bias:

- Coverage bias → Some members of the population have little or no chance of being included because the sampling method does not adequately cover the population.
- Selection bias → The method used to select participants systematically favors certain types of people over others.
- Nonresponse bias → People who do not respond differ systematically from those who do respond in a way related to the question being studied.

Coverage = Who can be reached	Selection = Who gets selected	Nonresponse = Who actually responds
-------------------------------	-------------------------------	-------------------------------------

Sampling Methods

Simple Random Sampling equal chance for everyone.

Stratified Sampling divide into groups, then sample from EACH group.

Cluster divide into groups, randomly choose SOME groups, then sample within those selected groups

Systematic every k -th unit after a random start

5.3 Sampling Distribution and the CLT

5.3.1 Sampling Distribution and SE

Suppose you drew a different random sample of 2,500 households and recomputed the average spend each time, 1000 times. What would the distributions of those averages be? How would that be different from the the distribution of the raw values themselves?

The sampling distribution is the distribution of a statistic (e.g., $\bar{X}$) across all possible repeated samples of the same size from a population.

Center tells us where the statistic tends to fall and whether it is centered on the true population parameter.

If $E(\bar{X}) = \mu$, $\bar{X}$ is an unbiased estimator of μ , the true value (in this case the mean).

Spread tells us the precision of the statistic across samples.

For sample mean: $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ where smaller SE (standard error) means less sample-to-sample variability.

Standard Error measures how much a statistic would typically vary from sample to sample.

5.3.2 Central Limit Theorem

The central limit theorem says as the sample size, n , gets larger, the sampling distribution of the sample mean, $\bar{X}$ becomes approximately normal, regardless of the shape of the population distribution.

*Rule of thumb, $n \geq 30$ is often “large enough” though more skew may require a larger n .

The CLT allows us to build **confidence intervals** and **hypothesis tests** even when the raw data isn't normal.

5.4 Inference for the Mean

When trying to estimate the mean of a population, we use a point estimate for the mean $\bar{X}$.

A point estimate is a single value calculated from a sample that is used to estimate an unknown population parameter.

When doing inference, the true standard deviation, σ , is unknown. Because of this, we use the **t-distribution** to model it (t_{n-1}).

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

Where:

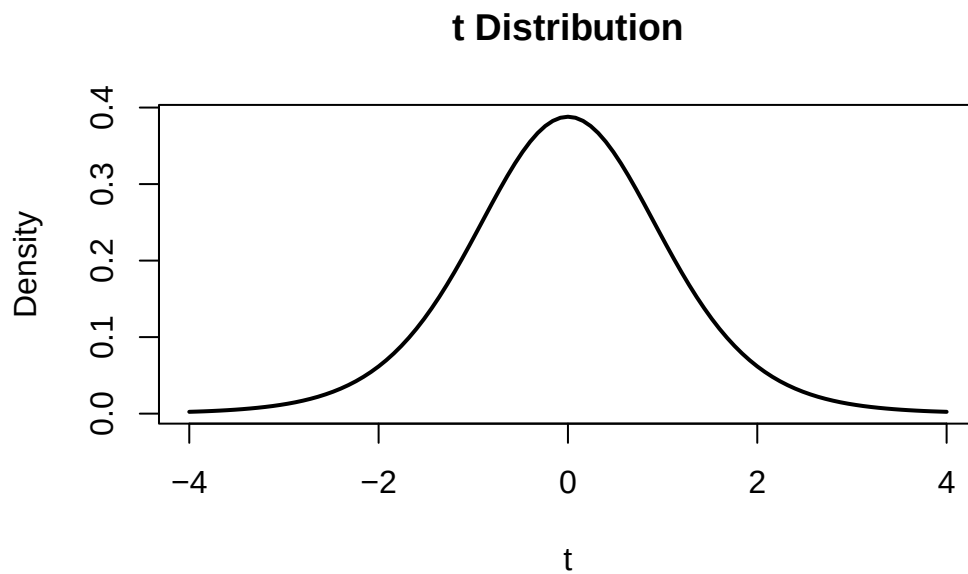
- $\bar{X} \rightarrow$ point estimate of
- $t_{\alpha/2, n-1} \rightarrow$ critical t-value with $n-1$ degrees of freedom (desired confidence level)
- $\frac{s}{\sqrt{n}} \rightarrow$ standard error
- $t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \rightarrow$ margin of error

Interpretation: the procedure covers μ in $100(1 - \alpha)$ of repeated samples. e.g. We are 95% confident that the mean is within...

```

curve(
  dt(x, df = 9),
  from = -4,
  to = 4,
  lwd = 2,
  xlab = "t",
  ylab = "Density",
  main = "t Distribution"
)

```



5.4.1 t vs z

A t-statistic is like a z-statistic when the population standard deviation σ is unknown and must be estimated using s .

	z	t
Distribution	Standard normal $N(0, 1)$	t -distribution
Population SD σ	Known	Unknown
Standard error	$\frac{\sigma}{\sqrt{n}}$	$\frac{s}{\sqrt{n}}$

	z	t
Degrees of freedom	None	$n - 1$
Shape	Normal	Normal-like, heavier tails
Critical value for 95%	1.96	Depends on df
Common use for means	Less common	Most common

As n gets larger, the difference between t and z becomes smaller and eventually negligible because s becomes a more precise estimate of σ .

Helpful Definitions:

Distribution a z -statistic follows the standard normal distribution $N(0, 1)$. A t -statistic follows a t -distribution.

Population SD σ if the true population SD σ is known, use z . If σ is unknown, estimate it with the sample SD s and use t .

Standard error measures how much $\bar{X}$ varies across samples. With z , $SE = \frac{\sigma}{\sqrt{n}}$. With t , because σ is unknown, we estimate it with $SE = \frac{s}{\sqrt{n}}$.

Degrees of freedom z doesn't need degrees of freedom because σ is known. t uses $df = n - 1$, which accounts for estimating variability using the sample.

Shape both are bell-shaped and centered at 0. The t -distribution has heavier tails because estimating σ with s introduces additional uncertainty.

Critical value for 95% for z , the middle 95% is always between -1.96 and +1.96; for t , it depends on n ($n - 1$).

Common use for means knowing the true population σ is uncommon, so z is less common for inference about means. Usually σ is unknown, making t the standard choice.

5.4.2 Hypothesis Testing for the Mean

$H_0 \rightarrow$ the Null Hypothesis: the baseline/default claim. $H_a \rightarrow$ the Alternative Hypothesis: what we are looking for evidence of (represents a difference/effect/change from null)

Steps: hypothesis $\rightarrow$ significance level α test statistic $\rightarrow p$ -value $\rightarrow$ decision $\rightarrow$ conclusion

- α = probability of a **Type I error** (rejecting a true H_0)
- β = probability of a **Type II error** (failing to reject a false H_0)
- Power = $1 - \beta$ is the probability that a hypothesis test correctly detects a real effect.

Error	What Happens	Probability	Think
Type I	Reject a true H_0	α	False positive
Type II	Fail to reject a false H_0	β	False negative

5.4.2.1 One-Sided t-Test and Power

A **one-sided t-test** tests whether a population mean is **larger than or smaller than** a hypothesized value, depending on the specified direction.

- Right-tailed: $H_A : \mu > \mu_0$
- Left-tailed: $H_A : \mu < \mu_0$

Reject H_0 when: $p\text{-value} \leq \alpha$

5.5 Inference for a Proportion

Inference for a proportion uses essentially the same inference framework as inference for a mean, but the **population parameter changes**.

For means, the population parameter is: μ = population mean

For proportions, the population parameter is: p = population proportion

Same inference process, new parameter.

	Mean μ	Proportion p
Point Estimate	$\bar{X}$	$\hat{p}$
Standard Error	$\frac{s}{\sqrt{n}}$	$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Reference Distribution	t_{n-1}	z (Normal)
Confidence Interval	$\bar{X} \pm t_{\alpha/2, n-1} \cdot SE$	$\hat{p} \pm z_{\alpha/2} \cdot SE$
Test Statistic (H_0)	$\frac{\bar{X} - \mu_0}{s/\sqrt{n}}$	$\frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$

For proportion inference, each observation is a **binary outcome**:

$$X_i = \begin{cases} 1 & \text{success} \\ 0 & \text{failure} \end{cases}$$

Each observation follows a Bernoulli distribution:

$X_i \sim \text{Bernoulli}(p)$ where p is the true population probability or proportion of success.

6 Comparing Two Population Means: Paired Samples

6.1 Overview

Lecture 5 introduced inference for **one population parameter**.

For example, we could ask whether a population mean differs from some benchmark:

$$H_0 : \mu = \mu_0$$

Now suppose we have **two sets of measurements** and want to compare their means.

Examples include:

- Customer spending before vs. after joining a loyalty program
- Store sales before vs. after a layout change
- Measurements from matched treatment and control stores
- Measurements from twins or other naturally matched subjects

Before choosing a statistical test, the most important question is:

Are the observations in the two conditions independent, or are they meaningfully connected?

If observations can be meaningfully linked one-to-one, we have **paired data**.

If no meaningful one-to-one relationship exists, we instead have **independent samples**.

This distinction determines how the data should be analyzed.

7 Paired vs. Independent Samples

Suppose a retailer measures checkout time at 20 stores.

The company then installs a new checkout layout and measures checkout time at the **same 20 stores again**.

For every store, there are two measurements:

$$(\text{Before}_i, \text{After}_i)$$

These observations are **paired** because each after measurement corresponds to the same store's before measurement.

It would be inappropriate to ignore this connection and treat the 40 measurements as unrelated observations.

Instead, calculate the change within each store:

$$d_i = \text{After}_i - \text{Before}_i$$

Now the original problem:

Before vs. After

becomes:

$$d_1, d_2, \dots, d_n$$

which is a **one-sample problem involving the differences**.

7.1 What Creates a Pair?

A pair exists when an observation in one condition can be meaningfully linked to exactly one observation in another condition.

Common forms of pairing include:

7.1.1 Repeated Measurements

The same observational unit is measured twice.

Examples:

- Same customer before and after a program
- Same employee before and after training
- Same store before and after a redesign

7.1.2 Natural Pairs

Different observational units have a meaningful natural relationship.

Examples:

- Identical twins
- Siblings
- Matched geographic locations

7.1.3 Matched Pairs

Researchers intentionally match similar observational units before comparing treatments.

Examples:

- Match two stores with similar historical sales
- Match customers with similar demographics
- Match patients with similar baseline characteristics

One member of each pair may then receive one treatment while the other receives another.

8 Why Pairing Matters

Suppose we have measurements:

$$X_i$$

and:

$$Y_i$$

for pair i .

Instead of analyzing the two sets separately, calculate:

$$d_i = X_i - Y_i$$

Each pair now contributes **one difference**.

The pair, rather than each individual measurement, becomes the **unit of analysis**.

Pairing can improve precision because characteristics shared within a pair are removed before the two conditions are compared.

For example, stores may naturally differ because of:

- Store size
- Customer traffic
- Location
- Neighborhood demographics
- Historical sales

If the same store is measured before and after a change, much of this store-to-store variation is automatically controlled because each store is compared with itself.

Conceptually:

$$\text{Observed Difference} = \text{Condition Effect} + \text{Within-Pair Noise}$$

rather than comparing two groups containing all of the between-unit variation.

When the measurements within pairs are meaningfully related, pairing can therefore reduce noise and improve precision.

9 Defining the Difference

The direction of the difference matters.

Suppose:

$$d_i = \text{After}_i - \text{Before}_i$$

Then:

- $d_i > 0$ means the measurement increased
- $d_i < 0$ means the measurement decreased
- $d_i = 0$ means there was no change

If the outcome is checkout time, for example:

$$d_i < 0$$

would represent an improvement because checkout time decreased.

But if we instead defined:

$$d_i = \text{Before}_i - \text{After}_i$$

then improvement would produce:

$$d_i > 0$$

There is nothing inherently correct about one direction over the other.

The important thing is to:

1. Define the difference explicitly
2. Keep the direction consistent
3. Interpret the sign in context

10 The Population Parameter

Once the differences are defined:

$$d_1, d_2, \dots, d_n$$

the parameter of interest is:

$$\boxed{\mu_d}$$

where:

$$\mu_d = \text{population mean difference}$$

The corresponding sample statistic is:

$$\boxed{\bar{d}}$$

where:

$$\bar{d} = \frac{\sum d_i}{n}$$

Therefore:

$$\bar{d} \rightarrow \text{estimate of } \mu_d$$

This is exactly the same logic used in one-sample inference:

$$\bar{x} \rightarrow \mu$$

except the observations are now **within-pair differences**.

11 Hypotheses for Paired Data

If the question is whether the two conditions differ on average, the null hypothesis is:

$$H_0 : \mu_d = 0$$

A mean difference of zero means:

There is no average difference between the two conditions.

The alternative depends on the research question.

11.0.1 Two-Sided

To test whether the conditions differ:

$$H_a : \mu_d \neq 0$$

11.0.2 Right-Tailed

If positive differences represent the effect of interest:

$$H_a : \mu_d > 0$$

11.0.3 Left-Tailed

If negative differences represent the effect of interest:

$$H_a : \mu_d < 0$$

The direction depends entirely on how:

$$d_i$$

was defined.

12 Paired t-Test

A **paired t-test** compares the population mean difference to a hypothesized value, usually zero.

The important insight is:

A paired t-test is mathematically a one-sample t-test performed on the within-pair differences.

Once we calculate:

$$d_i = X_i - Y_i$$

we no longer need to think of the analysis as two separate samples.

We have one sample:

$$d_1, d_2, \dots, d_n$$

and ask whether its population mean differs from zero.

12.1 Standard Error of the Mean Difference

Let:

$$s_d$$

represent the sample standard deviation of the differences.

Then:

$$SE(\bar{d}) = \frac{s_d}{\sqrt{n}}$$

where:

n = number of **pairs**

Notice that n is not the total number of individual measurements.

If there are 15 pairs, then:

$$n = 15$$

even though there are:

$$30$$

individual measurements.

12.2 Paired t-Test Statistic

To test:

$$H_0 : \mu_d = 0$$

the test statistic is:

$$t = \frac{\bar{d} - 0}{s_d / \sqrt{n}}$$

More generally:

$$t = \frac{\bar{d} - \mu_{d,0}}{s_d / \sqrt{n}}$$

The degrees of freedom are:

$$df = n - 1$$

The test statistic tells us:

How many standard errors the observed mean difference lies from the value specified by the null hypothesis.

A large absolute value of t means the observed mean difference is far from the null value relative to the amount of sampling uncertainty.

13 Example: Identical Twin Study

Suppose a study compares left hippocampal volume for **15 pairs of identical twins**.

Within each pair:

- One twin was diagnosed with schizophrenia
- The other twin was not

Define:

$$d_i = \text{Unaffected}_i - \text{Affected}_i$$

The observed differences have:

$$\bar{d} = 0.1987$$

and:

$$s_d = 0.2383$$

Because:

$$n = 15$$

the standard error is:

$$SE(\bar{d}) = \frac{0.2383}{\sqrt{15}}$$

$$SE(\bar{d}) \approx 0.0615$$

Under:

$$H_0 : \mu_d = 0$$

the test statistic is:

$$t = \frac{0.1987 - 0}{0.0615}$$

$$t \approx 3.23$$

with:

$$df = 15 - 1 = 14$$

So the observed mean difference is approximately:

$$3.23$$

standard errors above the null value of zero.

13.1 Paired t-Test in R

If the differences have already been calculated:

```
t.test(  
  pairs$Difference,  
  mu = 0  
)
```

This is simply a **one-sample t-test on the differences**.

The example produces approximately:

$$t = 3.23$$

$$df = 14$$

$$p = 0.006$$

and a 95% confidence interval of approximately:

$$(0.067, 0.331)$$

Because:

$$p < 0.05$$

we reject:

$$H_0 : \mu_d = 0$$

The data provide evidence that the population mean difference is not zero.

13.2 Using `paired = TRUE`

R can also calculate the differences internally.

For example:

```
t.test(  
  pairs$Unaffected,  
  pairs$Affected,  
  paired = TRUE  
)
```

This performs the same conceptual analysis:

$$d_i = \text{Unaffected}_i - \text{Affected}_i$$

followed by a one-sample t-test on those differences.

14 Confidence Interval for a Mean Difference

The confidence interval for the population mean difference follows the same structure as the confidence interval for a single population mean:

$$\bar{d} \pm t^* \frac{s_d}{\sqrt{n}}$$

For the twin example, the 95% confidence interval is approximately:

$$(0.067, 0.331)$$

Because the entire interval lies above zero, the data are consistent with a positive population mean difference under the chosen definition:

$$d_i = \text{Unaffected}_i - \text{Affected}_i$$

The confidence interval also communicates the estimated **magnitude** of the difference, rather than merely whether the difference is statistically significant.

15 Statistical Evidence vs. Causality

A statistically significant difference does **not automatically establish causation**.

In the twin example, the analysis provides evidence that mean left hippocampal volume differs between twins with and without a schizophrenia diagnosis.

That supports a statement about **association**:

Mean hippocampal volume differs between the two conditions.

It does not, by itself, establish:

Schizophrenia caused the difference in hippocampal volume.

Causal conclusions depend on the **study design**, not simply on obtaining a small p-value.

This connects back to the distinction between observational and experimental studies:

Statistical significance does not turn an observational study into a randomized experiment.

16 Matched-Pairs Experimental Design

Pairing can also be intentionally incorporated into an experiment.

Suppose Kroger has candidate stores for an endcap-display experiment.

Some stores will receive:

- **Treatment:** endcap display

while others receive:

- **Control:** usual shelf placement

Instead of randomly assigning stores without considering their characteristics, Kroger could use a **matched-pairs design**.

16.1 Step 1: Match Similar Units

Before treatment assignment, form pairs of similar stores using pre-experiment characteristics such as:

- Baseline sales
- Store size
- Customer traffic
- Neighborhood characteristics

The goal is to make the stores within each pair as comparable as possible before the treatment occurs.

16.2 Step 2: Randomize Within Each Pair

Within each matched pair:

- Randomly assign one store to treatment
- Randomly assign the other to control

Randomization remains important.

Matching improves comparability and precision, while random assignment supports causal inference.

16.3 Step 3: Calculate One Difference Per Pair

For pair i , calculate:

$$d_i = \text{Sales}_{\text{endcap},i} - \text{Sales}_{\text{control},i}$$

Then analyze:

$$d_1, d_2, \dots, d_n$$

and test:

$$H_0 : \mu_d = 0$$

The key distinction is:

Matching improves precision. Random assignment supports causality.

They serve different purposes.

17 Assumptions of the Paired t-Test

Once paired data are converted into differences:

$$(X_i, Y_i) \rightarrow d_i = X_i - Y_i$$

the assumptions of the paired t-test apply primarily to the **differences**.

The main assumptions are:

1. There is a meaningful reason for the observations to be paired
2. The differences are independent across pairs
3. The differences contain no severe outliers
4. The distribution of the differences is approximately Normal when the number of pairs is small

17.1 Independence Within a Pair Is Not Required

The two observations within a pair do **not** need to be independent.

In fact, the reason for using a paired analysis is usually that they are related.

For example:

$$(\text{Before}_i, \text{After}_i)$$

for the same store are naturally connected.

That relationship is exactly what the paired analysis preserves.

What needs to be independent is:

$$d_1, d_2, \dots, d_n$$

across separate pairs.

17.2 Normality Applies to the Differences

A paired t-test does **not** require both original variables to be Normally distributed.

You do not separately need:

$$X \sim N$$

and:

$$Y \sim N$$

Instead, the relevant quantity is:

$$D = X - Y$$

For small samples, we want the distribution of:

$$d_1, d_2, \dots, d_n$$

to be reasonably consistent with a Normal distribution and free of severe outliers.

Therefore:

Check the differences, not the two original columns separately.

18 Checking the Differences

After calculating:

```
pairs <- pairs |>
  mutate(
    Difference = Unaffected - Affected
  )
```

we can inspect the differences directly.

18.0.1 Histogram

```
ggplot(pairs, aes(x = Difference)) +
  geom_histogram()
```

Look for:

- Severe skewness
- Extreme outliers
- Unusual shape

18.0.2 Q-Q Plot

```
qqnorm(pairs$Difference)
qqline(pairs$Difference)
```

For a small sample, strong departures from the line may indicate that the Normality assumption is questionable.

However, with only a small number of pairs, visual diagnostics themselves are uncertain.

For example, with:

$$n = 15$$

there are relatively few observations available for judging the shape of the difference distribution.

19 Pairing Does Not Automatically Mean Independence

Paired data can introduce another important issue.

Suppose two stores are observed every week for 104 weeks.

For each week:

$$d_t = \text{Sales}_{8,t} - \text{Sales}_{208,t}$$

It might be tempting to treat the 104 weekly differences as:

$$d_1, d_2, \dots, d_{104}$$

independent observations.

But these differences occur **over time**.

Consecutive weeks may be related.

This creates a potential violation of the independence assumption.

20 Time Structure in Paired Differences

Suppose the difference between two stores changes systematically over time.

A single overall mean:

$$\bar{d}$$

might conceal that pattern.

For example, the differences might begin mostly negative and become increasingly positive.

Then the statement:

The average difference is 10.

does not fully describe what is happening.

The relationship may instead involve a **time trend**.

This connects paired inference back to time-series analysis:

Before reducing repeated measurements to one overall mean difference, check whether the difference itself changes systematically over time.

21 Autocorrelation

Autocorrelation measures whether observations are related to observations from earlier time periods.

For weekly differences:

$$d_t$$

we might ask whether:

$$d_t$$

is related to:

$$d_{t-1}$$

or:

$$d_{t-2}$$

and so on.

An **autocorrelation function (ACF)** examines these relationships at different lags.

For example:

```
acf(differences)
```

A bar at lag:

$$k$$

measures the relationship between observations separated by:

k

time periods.

If substantial autocorrelation exists, the observations do not provide completely independent pieces of information.

That matters because the usual paired t-test standard error:

$$SE(\bar{d}) = \frac{s_d}{\sqrt{n}}$$

assumes independent differences.

If observations are strongly dependent, treating all of them as independent can make the estimated uncertainty misleading.

22 Changing Means vs. Autocorrelation

Time-series diagnostics can reveal different problems.

A **time plot** can reveal whether the mean difference changes over time.

An **ACF** can reveal whether observations are serially dependent.

These are not the same issue.

For example:

- A time trend means the expected difference is changing
- Autocorrelation means observations at different times are related

A dataset can have:

- A changing mean with little autocorrelation
- Autocorrelation around a stable mean
- Both
- Neither

Therefore, both the **structure of the mean** and the **dependence between observations** matter.

23 Unit of Replication

One of the most important questions in statistical inference is:

What is the independent unit of replication?

Suppose we compare two study designs.

23.0.1 Study A: Many Matched Store Pairs

Suppose many pairs of similar stores participate in an experiment.

Each store is observed for six weeks.

We calculate one sales summary for each store and then one difference for each matched pair.

If there are:

$$n$$

independent store pairs, then the sample size is:

$$n$$

The independent replication comes from the **different store pairs**.

23.0.2 Study B: One Store Pair for 104 Weeks

Suppose instead that only two stores are studied, but they are observed for:

104

weeks.

There are 104 weekly differences, but they all come from the **same two stores**.

Those 104 weeks are not equivalent to having 104 independent store pairs.

They provide more information about those particular stores, but not necessarily more independent replication across a population of stores.

This distinction is critical:

More repeated observations of the same units are not the same as more independent observational units.

24 Pseudoreplication

Treating repeated or dependent measurements as though they were independent observations can create **pseudoreplication**.

For example:

Two stores observed for 104 weeks

does not automatically provide the same inferential information as:

104 independent pairs of stores.

Repeated observations may increase information about the particular units being studied, but they do not necessarily increase the number of independent units available for population-level inference.

The effective sample size depends on the study design and dependence structure, not simply the number of rows in the dataset.

25 Choosing Paired vs. Independent Analysis

Before comparing two means, ask:

Can each observation in one condition be meaningfully linked to one observation in the other condition?

25.0.1 If Yes

The observations are paired.

Calculate:

$$d_i = X_i - Y_i$$

and perform **one-sample inference on the differences**.

25.0.2 If No

The observations are independent samples.

Compare the two population means directly using an independent-samples method.

25.1 Examples

25.1.1 Same Customers Before and After

Monthly spending is recorded for the same customers before and after they join a loyalty program.

Paired

Each customer's before measurement is linked to that same customer's after measurement.

25.1.2 Randomized Website Experiment

One group of customers sees the current website.

A different group sees a redesigned website.

Independent

There is no one-to-one correspondence between customers in the two groups.

25.1.3 Matched Stores

Treatment stores are intentionally matched with similar control stores before an experiment.

Paired

Each treatment store has a specific matched control store.

25.1.4 Different Orders Across Months

Delivery times from one group of orders this month are compared with a different group of orders last month.

Independent

An order this month does not correspond to one specific order from last month.

26 Paired-Samples Workflow

When analyzing two conditions, use the following process:

1. **Identify the observational unit.**
2. Determine whether observations can be meaningfully paired.
3. If paired, define the within-pair difference:

$$d_i = X_i - Y_i$$

4. Determine the population parameter:

$$\mu_d$$

5. State the hypotheses:

$$H_0 : \mu_d = 0$$

and the appropriate alternative.

6. Check the distribution and independence of the differences.
7. Calculate:

$$\bar{d}$$

$$s_d$$

and:

$$SE(\bar{d}) = \frac{s_d}{\sqrt{n}}$$

8. Calculate:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

9. Obtain the p-value and confidence interval.
 10. Interpret the result in the context of the study.
 11. Make causal conclusions only if the study design supports them.
-

27 Overall Framework

The key idea is that a paired analysis preserves information about **which observations belong together**.

Instead of:

$$\text{Mean of Group 1} - \text{Mean of Group 2}$$

while ignoring the relationships between observations, paired analysis first calculates:

$$d_i = X_i - Y_i$$

for each pair.

The problem then becomes:

Two Paired Conditions $\rightarrow$ Within-Pair Differences $\rightarrow$ One-Sample Inference

The major ideas are:

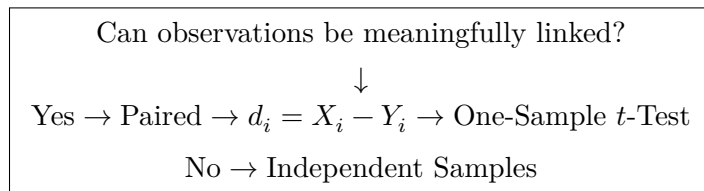
- **Study design determines the statistical test.**
- Pair observations only when there is a meaningful one-to-one connection.
- A paired t-test is a **one-sample t-test on differences**.
- The parameter of interest is the population mean difference:

$$\mu_d$$

- The sample size is the number of **independent pairs**, not the total number of measurements.
- Pairing can improve precision by removing variation between matched units.
- The assumptions apply primarily to the **differences**, not the two original columns.
- Independence is required **across pairs**, not within each pair.
- Repeated observations over time may introduce trends or dependence that a basic paired t-test does not capture.
- More repeated measurements are not equivalent to more independent units.

- Matching can improve precision, while **random assignment** is what supports a causal interpretation.

The decision rule for comparing two conditions is therefore:



28 Comparing Two Population Means: Independent Samples

28.1 Overview

Lecture 6 considered situations where observations from two conditions were **meaningfully linked**.

For paired data, each pair produced one difference:

$$d_i = X_{1i} - X_{2i}$$

and the analysis became a one-sample inference problem involving:

$$\mu_d$$

Now consider a different situation.

Suppose we want to compare:

- Spending by homeowners vs. renters
- Sales of national-brand vs. private-brand products
- Customers assigned to Website A vs. Website B
- Employees receiving one training program vs. employees receiving another

In these situations, the observations in one group do not have a meaningful one-to-one connection with observations in the other group.

These are **independent samples**.

Instead of analyzing within-pair differences, we compare the two sample means:

$$\bar{X}_1 - \bar{X}_2$$

to learn about the difference between the two population means:

$$\mu_1 - \mu_2$$

The major distinction is:

Paired samples compare within-pair differences. Independent samples compare two separate group means.

29 Why Study Design Still Matters

Statistical analysis involves more than calculating a p-value.

Suppose a retailer asks:

Would offering more coupons increase customer spending?

There are several different questions hidden inside this problem.

29.0.1 Describe

How does spending differ between customers who use coupons and customers who do not?

29.0.2 Quantify Uncertainty

How precisely can we estimate the difference in population mean spending?

29.0.3 Evaluate Causality

Does the difference actually result from coupons, or are customers who use coupons already different from customers who do not?

A hypothesis test addresses **sampling uncertainty**.

It does not automatically eliminate:

- Confounding
- Selection bias
- Differences in the underlying groups

Therefore:

Statistical significance and causal evidence are different questions.

The study design determines what conclusions the statistical analysis can support.

30 Paired vs. Independent Samples

Before comparing two means, determine whether the observations are **paired or independent**.

30.1 Paired Samples

Observations are paired when each observation in one condition has a meaningful one-to-one connection with an observation in the other condition.

Examples:

- Same customer before and after treatment
- Same store before and after a redesign
- Identical twins
- Matched treatment and control stores

Each pair produces:

$$d_i = X_{1i} - X_{2i}$$

and inference is performed on:

$$\mu_d$$

30.2 Independent Samples

Samples are independent when the groups contain **different observational units** and there is no meaningful one-to-one link between observations across groups.

Examples:

- 20 randomly selected men vs. 20 independently selected women
- Homeowners vs. renters
- National-brand products vs. private-brand products
- Customers assigned to two different website versions

Each observational unit contributes to only **one group**.

The parameter of interest is:

$$\mu_1 - \mu_2$$

31 The Observational Unit Determines the Design

The **observational unit** is the individual entity on which a measurement is taken.

Before choosing a statistical test, ask:

What is the observational unit?

and then:

Does each observational unit contribute one measurement or a linked pair of measurements?

If each unit contributes to only one group:

Independent Samples

If the same unit contributes measurements under both conditions, or units are intentionally matched:

Paired Samples

31.1 Example: Taste Test

Suppose a company wants to compare mean taste ratings for Coke and Pepsi.

31.1.1 Independent-Samples Design

Randomly assign participants to one of two groups:

- Group 1 tastes Coke
- Group 2 tastes Pepsi

Each participant tastes only one product.

The groups are independent.

31.1.2 Paired Design

Every participant tastes both Coke and Pepsi.

Each participant provides:

$$(\text{Coke Rating}_i, \text{Pepsi Rating}_i)$$

The observations are paired because both ratings come from the same person.

The two designs answer similar questions but use different information.

The paired design controls for individual differences in taste preferences because each participant acts as their own comparison.

However, the study would need to prevent **order effects**.

For example, tasting Coke first might affect a participant's Pepsi rating.

A common solution would be to randomize which product each participant tastes first.

32 Independent Samples and the Population Difference

Suppose Group 1 has population mean:

$$\mu_1$$

and Group 2 has population mean:

$$\mu_2$$

The parameter of interest is:

$$\Delta = \mu_1 - \mu_2$$

The sample estimate is:

$$\hat{\Delta} = \bar{X}_1 - \bar{X}_2$$

Therefore:

$$\bar{X}_1 - \bar{X}_2 \rightarrow \text{estimate of } \mu_1 - \mu_2$$

The direction must be defined **before interpreting the result**.

For example, suppose:

$$\Delta = \mu_{\text{Homeowners}} - \mu_{\text{Renters}}$$

Then:

- $\Delta > 0$ means homeowners spend more on average
- $\Delta < 0$ means renters spend more on average
- $\Delta = 0$ means the population means are equal

Reversing the order reverses the sign:

$$\mu_1 - \mu_2 = -(\mu_2 - \mu_1)$$

Therefore:

Always define the direction of the difference before interpreting it.

33 Hypotheses for Two Independent Means

The most common null hypothesis is:

$$H_0 : \mu_1 - \mu_2 = 0$$

which is equivalent to:

$$H_0 : \mu_1 = \mu_2$$

The alternative depends on the research question.

33.0.1 Two-Sided

To determine whether the population means differ:

$$H_a : \mu_1 - \mu_2 \neq 0$$

33.0.2 Right-Tailed

To determine whether Population 1 has a larger mean:

$$H_a : \mu_1 - \mu_2 > 0$$

33.0.3 Left-Tailed

To determine whether Population 1 has a smaller mean:

$$H_a : \mu_1 - \mu_2 < 0$$

Again, the meaning of positive and negative depends on how Group 1 and Group 2 were defined.

34 Two Samples, Two Sources of Uncertainty

With paired samples, uncertainty comes from the variation among the within-pair differences.

With independent samples, we estimate:

$$\mu_1$$

using:

$$\bar{X}_1$$

and independently estimate:

$$\mu_2$$

using:

$$\bar{X}_2$$

Both estimates contain sampling uncertainty.

Therefore:

$$\bar{X}_1 - \bar{X}_2$$

has uncertainty from **both samples**.

For independent samples:

$$SE(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

In practice, the population standard deviations are usually unknown, so they must be estimated from the samples.

An important consequence is:

The less precise group contributes more uncertainty to the comparison.

For example, if:

$$n_1 = 10,000$$

but:

$$n_2 = 20$$

the second sample will generally contribute much more uncertainty to the estimated difference.

A huge sample in one group does not eliminate uncertainty created by a small sample in the other.

35 Two-Sample t-Tests

There are two major approaches for comparing two independent population means:

1. **Welch's two-sample t-test**
2. **Pooled two-sample t-test**

Both estimate the same population quantity:

$$\mu_1 - \mu_2$$

The difference is how they estimate the uncertainty.

36 Welch's Two-Sample t-Test

Welch's t-test estimates the variance of each population separately.

It does **not** assume that the two populations have equal variances.

The estimated population difference is:

$$\hat{\Delta} = \bar{X}_1 - \bar{X}_2$$

The standard error is:

$$SE(\hat{\Delta}) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where:

s_1 = sample standard deviation of Group 1

s_2 = sample standard deviation of Group 2

n_1 = sample size of Group 1

n_2 = sample size of Group 2

The test statistic is:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

More generally, if the null hypothesis specifies some difference:

$$\Delta_0$$

then:

$$t = \frac{\hat{\Delta} - \Delta_0}{SE(\hat{\Delta})}$$

As before, the t-statistic represents:

How many estimated standard errors the observed difference lies from the null value.

36.1 Welch's Test in R

R uses Welch's method by default when performing an independent two-sample t-test.

```
t.test(  
  outcome ~ group,  
  data = my_data  
)
```

For example:

```
t.test(  
  spending ~ homeowner_desc,  
  data = households  
)
```

Unless otherwise specified, `t.test()` does **not** assume equal population variances.

This makes Welch's method the standard default when there is no strong reason to assume that the two population variances are equal.

37 Pooled Two-Sample t-Test

The **pooled t-test** makes an additional assumption:

$$\sigma_1^2 = \sigma_2^2 = \sigma^2$$

In other words, the two populations are assumed to have the **same variance**.

Because both samples are assumed to estimate the same population variance, their information can be combined into a single **pooled variance estimate**.

The pooled variance is:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

The pooled standard deviation is:

$$s_p = \sqrt{s_p^2}$$

The standard error becomes:

$$SE_{\text{pooled}} = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

The degrees of freedom are:

$$df = n_1 + n_2 - 2$$

37.1 Pooled Test in R

Unlike Welch's method, pooling must be requested explicitly:

```
t.test(  
  outcome ~ group,  
  data = my_data,  
  var.equal = TRUE  
)
```

The argument:

```
var.equal = TRUE
```

tells R to assume equal population variances.

38 Welch vs. Pooled t-Test

Both methods estimate:

$$\mu_1 - \mu_2$$

but they make different assumptions about variability.

Feature	Welch	Pooled
Compares two independent means	Yes	Yes
Assumes equal population variances	No	Yes
Estimates variances separately	Yes	No
Combines variance estimates	No	Yes
R <code>t.test()</code> default	Yes	No

The practical rule is:

Use Welch's method by default unless there is a defensible reason to assume equal population variances.

The pooled method is appropriate only when the equal-variance assumption has a reasonable basis.

39 Why Unequal Sample Sizes Matter

Unequal sample sizes are especially important when the groups also have different variances.

Recall Welch's standard error:

$$SE(\hat{\Delta}) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Each group's contribution depends on both:

- Its variability
- Its sample size

A group with:

- High variability
- Small sample size

can contribute substantial uncertainty.

Therefore, before comparing two means, examine:

$$n_1, n_2, s_1, s_2$$

rather than looking only at the two means.

40 Examine the Data Before Testing

Before running a two-sample t-test, summarize and visualize both groups.

Important features include:

- Sample size
- Mean
- Median
- Standard deviation
- Skewness
- Outliers or unusual observations

For example:

```
my_data |>
  group_by(group) |>
  summarise(
    n = n(),
    mean = mean(outcome, na.rm = TRUE),
    median = median(outcome, na.rm = TRUE),
    sd = sd(outcome, na.rm = TRUE)
  )
```

A useful visualization is a boxplot:

```
ggplot(
  my_data,
  aes(
    x = group,
    y = outcome
  )
) +
  geom_boxplot()
```

The statistical test should not replace descriptive analysis.

Instead:

Describe $\rightarrow$ Check $\rightarrow$ Infer

41 Example: National vs. Private-Brand Soft Drinks

Suppose we compare total observed sales for:

- National-brand 2-liter soft drinks
- Private-brand 2-liter soft drinks

The observational unit is a:

Soft-drink product

Each product belongs to either:

- National brand

or:

- Private brand

A national-brand product is not meaningfully paired with one particular private-brand product.

Therefore, the samples are:

Independent

41.1 Descriptive Statistics

Suppose the data show:

Brand	Products	Mean	Median	SD
National	175	203.9	4.5	664.7
Private	57	170.4	62.1	263.9

Several features immediately deserve attention.

41.1.1 Unequal Sample Sizes

$$n_{\text{National}} = 175$$

while:

$$n_{\text{Private}} = 57$$

41.1.2 Unequal Variability

$$s_{\text{National}} = 664.7$$

while:

$$s_{\text{Private}} = 263.9$$

The national-brand group is much more variable.

41.1.3 Strong Skewness

For national brands:

$$\text{Mean} = 203.9$$

while:

$$\text{Median} = 4.5$$

The enormous difference between the mean and median indicates a highly right-skewed distribution with some extremely large observations.

This is why descriptive analysis should occur **before** inferential analysis.

42 Applying Welch's Test

Define the population difference as:

$$\Delta = \mu_{\text{National}} - \mu_{\text{Private}}$$

Suppose the estimated means are:

$$\bar{X}_{\text{National}} = 203.89$$

and:

$$\bar{X}_{\text{Private}} = 170.41$$

Then:

$$\hat{\Delta} = 203.89 - 170.41$$

$$\boxed{\hat{\Delta} = 33.48}$$

So national-brand products have an observed mean sales value approximately 33.48 units higher than private-brand products.

But the estimate alone does not tell us how uncertain that difference is.

43 Confidence Interval for Two Independent Means

A confidence interval for the population mean difference has the general form:

$$\hat{\Delta} \pm t^* SE(\hat{\Delta})$$

For Welch's method:

$$(\bar{X}_1 - \bar{X}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Suppose the soft-drink example produces:

$$95\% \text{ CI} = (-87.14, 154.10)$$

This means the data are compatible with population mean differences ranging from approximately:

$$-87.14$$

to:

$$154.10$$

under the method's assumptions.

Because the interval contains:

$$0$$

a population mean difference of zero remains compatible with the data.

44 Confidence Intervals and Hypothesis Tests

For the two-sided hypotheses:

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_a : \mu_1 - \mu_2 \neq 0$$

the confidence interval and hypothesis test communicate closely related information.

44.0.1 If the 95% Confidence Interval Excludes 0

There is evidence against:

$$H_0 : \mu_1 - \mu_2 = 0$$

at the corresponding 5% significance level.

44.0.2 If the 95% Confidence Interval Contains 0

We fail to reject:

$$H_0 : \mu_1 - \mu_2 = 0$$

at the corresponding 5% significance level.

For the soft-drink example:

$$(-87.14, 154.10)$$

contains:

$$0$$

so we fail to reject the null hypothesis.

45 Failing to Reject Does Not Prove Equality

This distinction is essential.

Suppose:

$$p = 0.58$$

and:

$$95\% \text{ CI} = (-87.14, 154.10)$$

We can say:

The data do not provide sufficient evidence that the two population means differ.

We should **not** say:

The population means are equal.

Why?

Because the confidence interval contains many possible differences besides zero.

For example, values such as:

$$-50, 25, 100$$

are also inside the interval.

The data are therefore compatible with:

- Negative differences
- No difference
- Positive differences

The analysis has not demonstrated equality.

Failing to find evidence of a difference is not the same as finding evidence of no difference.

46 Statistical Significance vs. Practical Importance

Suppose a test finds:

$$p < 0.05$$

That tells us the observed difference would be difficult to explain purely through sampling variation under the null model.

It does **not** tell us whether the difference is important for the business.

For example, a very large sample might detect:

$$\$0.03$$

of additional average spending.

That difference might be statistically detectable but operationally irrelevant.

Therefore, interpretation should consider:

1. Estimated difference
2. Confidence interval
3. Statistical evidence
4. Practical magnitude
5. Study design

A p-value alone does not answer all of these questions.

47 Statistical Evidence vs. Causal Evidence

Suppose observational data show:

$$\bar{X}_{\text{Coupon}} > \bar{X}_{\text{No Coupon}}$$

Even if a two-sample t-test finds a statistically significant difference, we cannot automatically conclude:

Coupons caused customers to spend more.

Customers who use coupons may systematically differ from customers who do not.

For example, coupon users might:

- Shop more frequently
- Purchase larger baskets
- Be more loyal customers
- Differ demographically

A hypothesis test quantifies **sampling uncertainty**.

It does not automatically remove **confounding**.

Therefore:

Inference about a difference and inference about causation are separate issues.

48 Example: Homeowners vs. Renters

Suppose the business question is:

How does annual Kroger spending differ between households that own their home and households that rent?

First define the populations.

Population 1:

Kroger households that own their home

Population 2:

Kroger households that rent their home

Define:

$$\Delta = \mu_{\text{Owner}} - \mu_{\text{Renter}}$$

The observational unit is:

Household

Each household belongs to one group.

There is no natural one-to-one relationship between an owner household and a renter household.

Therefore:

Independent Samples

48.1 Analysis Workflow in R

First summarize the groups:

```
households |>
  group_by(homeowner_desc) |>
  summarise(
    n = n(),
    mean_spend = mean(spend, na.rm = TRUE),
    median_spend = median(spend, na.rm = TRUE),
    sd_spend = sd(spend, na.rm = TRUE)
  )
```

Then visualize:

```
ggplot(
  households,
  aes(
    x = homeowner_desc,
    y = spend
  )
) +
  geom_boxplot()
```

Then perform Welch's two-sample t-test:

```
t.test(
  spend ~ homeowner_desc,
  data = households
)
```

The output provides:

- Group means
- Estimated difference
- t-statistic
- Degrees of freedom
- p-value
- Confidence interval

The final interpretation should report the estimated population difference and its uncertainty in the context of homeowner and renter households.

49 Paired vs. Independent t-Tests

The two procedures are closely related but solve different problems.

Feature	Paired t-Test	Independent t-Test
Groups	Meaningfully linked	Separate
Observational units	Same/matched units	Different units
Parameter	μ_d	$\mu_1 - \mu_2$
Statistic	$\bar{d}$	$\bar{X}_1 - \bar{X}_2$
Source of uncertainty	Variation in differences	Variation from both samples
Analysis	One-sample t-test on differences	Two-sample t-test
Typical R argument	<code>paired = TRUE</code>	Welch default

The first question should therefore **not** be:

Which t-test formula do I use?

It should be:

How were the data generated, and are these observations meaningfully linked?

50 Independent-Samples Assumptions

For a two-sample t procedure, the analysis requires the observations to have an appropriate independence structure.

Most importantly:

50.0.1 Independence Between Groups

The observations in one group should not be paired with observations in the other group.

If meaningful pairing exists, use a paired analysis instead.

50.0.2 Independence Within Groups

Observations should represent independent observational units rather than repeated or clustered measurements being incorrectly treated as separate independent observations.

50.0.3 Distribution Shape

The behavior of the t procedure also depends on sample size and the distributions of the data.

Before inference, inspect:

- Skewness
- Extreme observations
- Sample sizes
- Variability

Large samples generally make inference about means more robust to moderate non-Normality, but extreme skewness and unusual observations should still be investigated.

50.0.4 Variance Assumption

Welch's test does **not** require:

$$\sigma_1^2 = \sigma_2^2$$

The pooled test does.

This is one reason Welch's method is generally preferred as the default.

51 Independent-Means Analysis: Full Workflow

When comparing two population means from separate groups:

51.1 Step 1: Identify the Observational Unit

Ask:

What does one row or measurement represent?

Determine whether the two groups contain genuinely separate observational units.

51.2 Step 2: Determine Whether the Samples Are Paired or Independent

Ask:

Can each observation in one group be meaningfully linked to exactly one observation in the other?

If yes:

Paired

If no:

Independent

51.3 Step 3: Define the Population Difference

Explicitly define:

$$\Delta = \mu_1 - \mu_2$$

before calculating or interpreting the estimate.

This establishes what positive and negative values mean.

51.4 Step 4: Examine the Data

For each group, examine:

- Sample size
- Mean
- Median
- Standard deviation
- Distribution shape
- Skewness
- Unusual observations

Use both numerical summaries and visualizations.

51.5 Step 5: Estimate the Difference

Calculate:

$$\hat{\Delta} = \bar{X}_1 - \bar{X}_2$$

51.6 Step 6: Quantify the Uncertainty

Using Welch's method:

$$SE(\hat{\Delta}) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Construct a confidence interval:

$$\hat{\Delta} \pm t^* SE(\hat{\Delta})$$

51.7 Step 7: Perform the Hypothesis Test

Usually:

$$H_0 : \mu_1 - \mu_2 = 0$$

against an appropriate alternative.

Use Welch's method by default:

```
t.test(  
  outcome ~ group,  
  data = my_data  
)
```

Use the pooled method only when the equal-population-variance assumption has a defensible basis:

```
t.test(  
  outcome ~ group,  
  data = my_data,  
  var.equal = TRUE  
)
```

51.8 Step 8: Interpret in Context

Report:

1. The direction of the comparison
2. The estimated mean difference
3. The confidence interval
4. The statistical evidence
5. Practical importance when relevant
6. Whether the study design supports a causal interpretation

The statistical result should always be translated back into the original business question.

52 Overall Framework

The sequence from Lectures 5–7 can now be viewed as one unified framework.

52.0.1 One Population Mean

Parameter:

$$\mu$$

Estimate:

$$\bar{X}$$

Method:

One-Sample t

52.0.2 Two Linked Conditions

Parameter:

$$\mu_d$$

Estimate:

$$\bar{d}$$

Method:

Paired t = One-Sample t on Differences

52.0.3 Two Independent Populations

Parameter:

$$\mu_1 - \mu_2$$

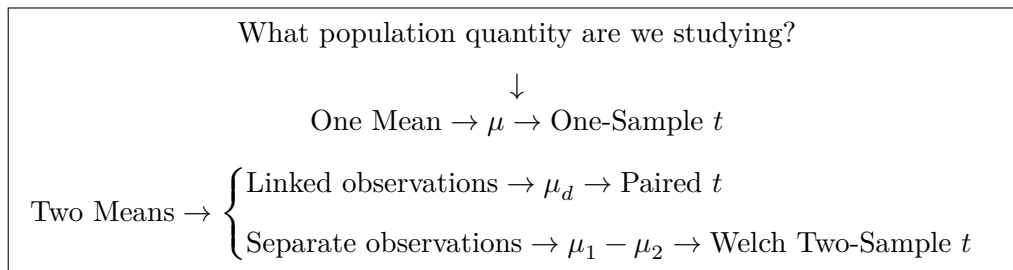
Estimate:

$$\bar{X}_1 - \bar{X}_2$$

Method:

Welch Two-Sample t

The decision process is therefore:



The key principle is:

The study design determines the statistical method.

Do not begin by choosing a formula.

Begin by identifying:

1. The business question
2. The observational unit
3. The populations being compared
4. Whether observations are linked or independent
5. The population parameter being estimated

Then choose the statistical method that matches that structure.

53 Inference for Categorical Data

53.1 Overview

Previous inference methods focused primarily on **quantitative outcomes**.

For example:

- One population mean $\rightarrow \mu$
- Paired mean difference $\rightarrow \mu_d$
- Difference between two independent means $\rightarrow \mu_1 - \mu_2$

Lecture 8 shifts the focus to **categorical outcomes**.

Examples include:

- Purchase vs. no purchase
- Homeowner vs. renter
- Married vs. unmarried
- Store vs. website vs. app
- Current page vs. new page vs. third page

Instead of asking questions about **means**, we now ask questions about:

- **Proportions**
- **Counts**
- **Associations between categorical variables**
- **Whether observed category frequencies match an expected distribution**

The main methods are:

1. **Two-proportion inference**
2. **Chi-square test of independence**
3. **Chi-square goodness-of-fit test**

The appropriate method depends on the business question and structure of the categorical data.

54 Two-Proportion Inference

Suppose a retailer conducts an A/B experiment.

Each visitor is randomly assigned to either:

- Current page
- New page

The outcome is whether the visitor:

- Purchases
- Does not purchase

Suppose the results are:

Page	Visitors	Purchases	Conversion Rate
Current	10,000	1,200	12.0%
New	10,000	1,230	12.3%

The sample proportions are:

$$\hat{p}_{Current} = \frac{1200}{10000} = 0.120$$

and:

$$\hat{p}_{New} = \frac{1230}{10000} = 0.123$$

The estimated difference is:

$$\hat{p}_{New} - \hat{p}_{Current} = 0.123 - 0.120 = 0.003$$

or:

0.3 percentage points

Notice that this is **0.3 percentage points**, not 0.3%.

54.1 Population Parameter

The population parameter is:

$$p_{New} - p_{Current}$$

where:

p_{New} = population conversion rate for the new page

$p_{Current}$ = population conversion rate for the current page

The sample estimate is:

$$\hat{p}_{New} - \hat{p}_{Current}$$

This follows the same general inference framework used previously:

Sample Statistic $\rightarrow$ Estimate Population Parameter

Here:

$$\hat{p}_{New} - \hat{p}_{Current} \rightarrow p_{New} - p_{Current}$$

55 Confidence Interval for Two Proportions

A confidence interval estimates the possible values of:

$$p_1 - p_2$$

using the observed sample proportions.

The standard error is:

$$SE = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

A 95% confidence interval is approximately:

$$(\hat{p}_1 - \hat{p}_2) \pm 1.96(SE)$$

For the page experiment, the estimated difference is:

0.30 percentage points

and the 95% confidence interval is:

$$(-0.61, 1.21) \text{ percentage points}$$

This interval includes:

$$0$$

so a population difference of zero is compatible with the data.

The interval also tells us much more than simply whether the result is statistically significant.

It describes the **range of plausible population differences**.

56 Practical Importance

Suppose the retailer decides that implementing the new page would only be worthwhile if conversion improves by at least:

0.5 percentage points

The confidence interval is:

$(-0.61, 1.21)$

This interval includes:

- Negative effects
- No effect
- Improvements smaller than 0.5 percentage points
- Improvements larger than 0.5 percentage points

Therefore, the data do not precisely establish whether the improvement reaches the business threshold.

This illustrates an important distinction:

Statistical uncertainty and business importance are separate questions.

A confidence interval allows us to compare the estimated effect with a practically meaningful threshold rather than looking only at whether:

$$p < 0.05$$

57 Hypothesis Test for Two Proportions

To test whether the population conversion rates differ:

$$H_0 : p_{New} = p_{Current}$$

which is equivalent to:

$$H_0 : p_{New} - p_{Current} = 0$$

Against the two-sided alternative:

$$H_a : p_{New} \neq p_{Current}$$

In R:

```
ab_test <- prop.test(  
  x = c(1230, 1200),  
  n = c(10000, 10000),  
  alternative = "two.sided",  
  correct = FALSE  
)  
  
ab_test
```

Here:

- `x` contains the number of successes
- `n` contains the total number of observations
- `alternative = "two.sided"` tests for any difference
- `correct = FALSE` disables the continuity correction

For the example:

$$p\text{-value} = 0.516$$

Because:

$$0.516 > 0.05$$

we fail to reject:

$$H_0$$

The experiment does not provide sufficient evidence that the population conversion rates differ.

This does **not** prove:

$$p_{New} = p_{Current}$$

It means the observed data do not provide sufficient evidence of a difference.

58 Why Proportions Are Pooled for the Hypothesis Test

There is an important difference between constructing a confidence interval and conducting the hypothesis test.

58.1 Confidence Interval

When constructing the confidence interval, we are **not assuming the population proportions are equal**.

Therefore, we estimate their variability separately using:

$$\hat{p}_1$$

and:

$$\hat{p}_2$$

The standard error is:

$$SE_{CI} = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

58.2 Hypothesis Test

Under the null hypothesis:

$$H_0 : p_1 = p_2$$

we are explicitly assuming that both groups share **one common population proportion**.

Therefore, we estimate that common proportion using the pooled sample proportion:

$$\hat{p}_{pool} = \frac{x_1 + x_2}{n_1 + n_2}$$

The hypothesis-test standard error uses:

$$\hat{p}_{pool}$$

because the test asks:

If the null hypothesis of equal population proportions were true, how much difference between the sample proportions would we expect from sampling variation alone?

So:

Confidence interval $\rightarrow$ keep proportions separate.

Hypothesis test under $H_0 : p_1 = p_2 \rightarrow$ pool the proportions.

59 From Two Groups to Several Groups

Suppose the retailer introduces a third page.

The results are now:

Page	Visitors	Purchases	Conversion Rate
Current	10,000	1,200	12.0%
New	10,000	1,230	12.3%
Third	15,000	2,100	14.0%

Now the question becomes:

Do conversion rates differ across the three pages?

The null hypothesis is:

$$H_0 : p_{Current} = p_{New} = p_{Third}$$

The alternative is:

$$H_a : \text{Not all population conversion rates are equal}$$

With more than two groups, we can organize the data into a **contingency table** and use a **chi-square test**.

60 Contingency Tables

A **contingency table**, also called a **two-way table**, displays the counts corresponding to combinations of two categorical variables.

For the page experiment, the two variables are:

1. Page
2. Purchase outcome

The table is:

Page	Purchased	Did Not Purchase
Current	1,200	8,800
New	1,230	8,770
Third	2,100	12,900

In R:

```
page_table <- cbind(  
  Purchased = pages$Purchases,  
  `Did not purchase` = pages$Visitors - pages$Purchases  
)  
  
rownames(page_table) <- pages$Page  
  
page_table
```

Each visitor contributes to exactly **one cell** in the table.

61 Why Raw Counts Alone Can Be Misleading

Notice that the third page has:

2100

purchases compared with:

1200

for the current page.

But the third page also has:

15000

visitors instead of:

10000

Therefore, comparing raw purchase counts alone would be misleading.

The relevant quantities are the **conditional proportions**:

$$\frac{\text{Purchases}}{\text{Visitors}}$$

which give:

- Current: 12.0%
- New: 12.3%
- Third: 14.0%

When group sizes differ, counts must be interpreted relative to the corresponding totals.

62 Chi-Square Test of Independence

The **chi-square test of independence** determines whether two categorical variables are associated.

For the page experiment, the variables are:

- Page assignment
- Purchase outcome

The null hypothesis is:

$$H_0 : \text{Page and purchase outcome are independent}$$

This can equivalently be written as:

$$H_0 : p_{Current} = p_{New} = p_{Third}$$

The alternative is:

$$H_a : \text{Page and purchase outcome are associated}$$

or equivalently:

$$H_a : \text{Not all population conversion rates are equal}$$

63 What Does Independence Mean?

Two categorical variables are **independent** if knowing the value of one does not change the distribution of the other.

In this example:

Page and purchase outcome are independent if knowing which page a visitor sees does not change their probability of purchasing.

If:

$$P(\text{Purchase} \mid \text{Current}) = P(\text{Purchase} \mid \text{New}) = P(\text{Purchase} \mid \text{Third})$$

then page and purchase are independent.

If these probabilities differ, the variables are associated.

64 Observed vs. Expected Counts

The chi-square test compares:

- **Observed counts** — what actually occurred
- **Expected counts** — what we would expect if the null hypothesis of independence were true

The observed counts are denoted:

$$O_{ij}$$

and the expected counts are denoted:

$$E_{ij}$$

The core question is:

If these variables were really independent, what would this table approximately look like?

65 Expected Counts Under Independence

Across all three pages:

$$1200 + 1230 + 2100 = 4530$$

visitors purchased.

There are:

$$10000 + 10000 + 15000 = 35000$$

total visitors.

Therefore, the overall purchase rate is:

$$\frac{4530}{35000} \approx 0.1294$$

or:

$$12.94\%$$

If page and purchase outcome are independent, each page should have approximately this same conversion rate.

For the current page:

$$E_{\text{Current,Purchased}} = 10000(0.1294) \approx 1294.3$$

More generally, expected counts are calculated as:

$$E_{ij} = \frac{(\text{Row Total}_i)(\text{Column Total}_j)}{\text{Grand Total}}$$

This formula preserves the observed row and column totals while constructing the table we would expect under independence.

66 Why Expected Counts Can Differ Across Groups

Independence does **not** mean that every group should have identical counts.

The third page has:

15000

visitors while the other pages have:

10000

each.

Even if all three pages have exactly the same underlying conversion probability, we would expect more purchases from the third page simply because more people saw it.

Therefore:

Independence implies comparable proportions, not necessarily identical counts.

67 Pearson Chi-Square Statistic

The Pearson chi-square statistic measures how far the observed counts are from the counts expected under the null hypothesis.

$$X^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Each cell contributes:

$$\frac{(O - E)^2}{E}$$

to the total statistic.

The logic is:

67.0.1 $O - E$

Measures how far the observed count is from the expected count.

67.0.2 Squaring

$$(O - E)^2$$

prevents positive and negative differences from canceling each other out.

67.0.3 Dividing by E

$$\frac{(O - E)^2}{E}$$

scales the discrepancy relative to the size of the expected count.

Therefore:

Larger X^2 values indicate greater disagreement between the observed data and the null model.

68 Degrees of Freedom

For a contingency table with:

- r rows
- c columns

the degrees of freedom are:

$$df = (r - 1)(c - 1)$$

For the page experiment:

$$r = 3$$

and:

$$c = 2$$

so:

$$df = (3 - 1)(2 - 1) = 2$$

Therefore:

$$df = 2$$

Under the null hypothesis and appropriate conditions:

$$X^2 \sim \chi^2_{(r-1)(c-1)}$$

approximately.

69 Chi-Square Test in R

The test can be conducted using:

```
page_test <- chisq.test(  
  page_table,  
  correct = FALSE  
)  
  
page_test
```

For the three-page example:

$$X^2 = 26.435$$

with:

$$df = 2$$

and:

$$p\text{-value} = 1.819 \times 10^{-6}$$

The p-value is extremely small, so we reject:

$$H_0$$

and conclude that page assignment and purchase outcome are associated.

Equivalently:

There is evidence that at least two of the population conversion rates differ.

70 What an Overall Chi-Square Test Does Not Tell You

A significant chi-square test tells us:

Some association exists

or:

Not all population proportions are equal

It does **not** automatically tell us:

- Every group differs from every other group
- Which specific groups differ
- Which group should be selected by the business
- Whether the differences are practically important

For the page example, rejecting:

$$H_0 : p_{Current} = p_{New} = p_{Third}$$

means the equality of all three proportions is inconsistent with the data.

It does **not** mean:

$$p_{Current} \neq p_{New} \neq p_{Third}$$

in every pairwise comparison.

The next step is to investigate **where the association appears**.

71 Two-Proportion Test and Chi-Square Connection

With exactly two groups and two outcomes, the contingency table is:

$$2 \times 2$$

For example:

Page	Purchased	Did Not Purchase
New	1230	8770
Current	1200	8800

The chi-square test can be run using:

```
two_page_table <- page_table[
  c("New", "Current"),
]

two_page_test <- chisq.test(
  two_page_table,
  correct = FALSE
)
```

For a two-sided comparison without continuity correction, the two-proportion test and chi-square test are mathematically connected:

$$X^2 = z^2$$

Therefore, they produce the same p-value in this setting.

For this example:

$$p \approx 0.516$$

from either method.

So the two-proportion test and chi-square test are not unrelated techniques.

The two-proportion test is closely connected to the chi-square framework when the table is:

$$2 \times 2$$

72 Finding Where the Association Occurs

After obtaining a significant overall chi-square test, first examine the category proportions.

For the page experiment:

Page	Conversion Rate
Current	12.0%
New	12.3%
Third	14.0%

The pooled conversion rate is approximately:

12.94%

The third page appears noticeably higher than the other two.

This provides a descriptive indication of where the association may be occurring.

However, we can investigate individual cells more formally using **standardized residuals**.

73 Standardized Residuals

A **standardized residual** measures how far an observed cell count differs from what would be expected under independence, accounting for the cell's variability.

In R:

```
round(
  page_test$stdres,
  2
)
```

For the page example:

Page	Purchased	Did Not Purchase
Current	-3.32	3.32
New	-2.27	2.27
Third	5.10	-5.10

Interpretation:

- **Positive residual** → more observations than expected
- **Negative residual** → fewer observations than expected
- **Large absolute residual** → stronger departure from independence

For the third page:

5.10

for purchases means substantially **more purchases occurred than expected under independence**.

Its non-purchase residual is:

−5.10

meaning substantially **fewer non-purchases occurred than expected**.
This pattern helps explain the significant overall chi-square statistic.

74 Why Residuals Come in Opposite Directions

Within a row, the total number of visitors is fixed.

If a page has more purchases than expected, those additional observations must come from somewhere.

Therefore, it will also have fewer non-purchases than expected.

For example:

$$O_{\text{Purchased}} > E_{\text{Purchased}}$$

will correspond with:

$$O_{\text{Not Purchased}} < E_{\text{Not Purchased}}$$

when there are only two outcome categories.

75 Conditions for the Chi-Square Approximation

Before trusting a chi-square test, check whether its assumptions are reasonable.

75.1 Independent Observations

Observations should contribute independently.

For example, each visitor should contribute one independent observation rather than the same visitor appearing repeatedly and being treated as independent observations.

A huge dataset does not solve a violation of independence.

More rows do not fix dependence among observations.

75.2 Mutually Exclusive Categories

Each observational unit should contribute to exactly one cell.

For example, a visitor cannot simultaneously be classified as both:

- Purchased
- Did not purchase

The categories must be mutually exclusive for the table being analyzed.

75.3 Expected Counts

Expected counts should be sufficiently large for the chi-square approximation to be reliable.

A simple conservative rule is:

$$E_{ij} \geq 5$$

for every cell.

Expected counts can be inspected in R using:

```
round(  
  page_test$expected,  
  1  
)
```

For the page example:

Page	Purchased	Did Not Purchase
Current	1294.3	8705.7
New	1294.3	8705.7
Third	1941.4	13058.6

All expected counts are far greater than 5.

76 Chi-Square Test of Independence: Workflow

When determining whether two categorical variables are associated:

76.1 Step 1: Identify the Observational Unit

Determine what one observation represents.

76.2 Step 2: Identify the Two Categorical Variables

For example:

- Page
 - Purchase outcome
-

76.3 Step 3: Construct a Contingency Table

```
table(  
  Variable1 = data$variable1,  
  Variable2 = data$variable2  
)
```

76.4 Step 4: Examine Counts and Proportions

Do not rely only on raw counts.

Compare relevant percentages or conditional proportions.

76.5 Step 5: State the Hypotheses

H_0 : The variables are independent

H_a : The variables are associated

76.6 Step 6: Calculate Expected Counts

Conceptually:

$$E_{ij} = \frac{(\text{Row Total})(\text{Column Total})}{\text{Grand Total}}$$

76.7 Step 7: Check Conditions

Verify:

- Independent observations
 - Mutually exclusive categories
 - Adequate expected counts
-

76.8 Step 8: Conduct the Test

```
test <- chisq.test(  
  contingency_table,  
  correct = FALSE  
)
```

76.9 Step 9: Interpret the Overall Result

If the p-value is small:

There is evidence of an association between the categorical variables.

If the p-value is not small:

The data do not provide sufficient evidence of an association.

76.10 Step 10: Investigate the Pattern

If the overall test is significant, examine:

- Conditional proportions
- Observed vs. expected counts
- Standardized residuals

```
test$expected  
test$stdres
```

This helps determine which categories appear to contribute most strongly to the overall association.

77 Example: Homeownership and Marital Status

Suppose we ask:

Is homeownership associated with marital status among the households in these data?

The two categorical variables are:

77.0.1 Homeownership

Possible categories include:

- Homeowner
- Probable Owner
- Probable Renter
- Renter
- Unknown

77.0.2 Marital Status

The codes are:

- A = Married
- B = Unmarried
- U = Unknown

A contingency table might be created using:

```
table(  
  Homeownership = hh$homeowner_desc,  
  Marital_status = hh$marital_status_code  
)
```

The hypotheses are:

H_0 : Homeownership and marital status are independent

H_a : Homeownership and marital status are associated

Before running the test, however, we must decide which categories should be included.

78 Category Definitions Matter

Suppose the data contain categories such as:

- Unknown homeownership
- Unknown marital status
- Probable owner
- Probable renter

The decision to include, combine, or exclude these categories changes the population represented by the analysis and therefore changes the interpretation.

For example, removing unknown marital status means the analysis is no longer about:

All households in the dataset.

It becomes an analysis of:

Households with known marital status.

Therefore:

Data-cleaning decisions are part of the statistical analysis because they determine what population the resulting inference describes.

Categories should not be removed simply because they are inconvenient.

The decision should follow from the research question and the meaning of the categories.

79 Association Does Not Imply Causation

Suppose the chi-square analysis finds an association between:

- Marital status
- Homeownership

Can we conclude:

Getting married causes people to become homeowners?

No.

The chi-square test establishes statistical evidence of an **association**.

It does not establish the causal mechanism behind that association.

Other variables might affect both marital status and homeownership, such as:

- Age
- Income
- Household size
- Employment
- Geographic location

Therefore:

Association $\neq$ Causation

The study design determines whether causal conclusions are justified.

80 Chi-Square Goodness-of-Fit Test

The chi-square framework can also answer a different type of categorical question.

Suppose a retailer expects customers to make their first purchase through:

- Store: 50%
- Website: 30%
- App: 20%

These percentages represent a **specified benchmark distribution**.

An independent sample of 500 customers produces:

Channel	Observed Customers	Benchmark Share
Store	240	50%
Website	160	30%
App	100	20%

The question is:

Does the observed channel mix match the retailer's specified benchmark distribution?

This is a **chi-square goodness-of-fit test**.

81 Independence vs. Goodness-of-Fit

These two chi-square procedures answer different questions.

81.1 Chi-Square Test of Independence

Uses **two categorical variables**.

Example:

Is page assignment associated with purchase outcome?

The data form a two-way contingency table.

81.2 Chi-Square Goodness-of-Fit Test

Uses **one categorical variable** and compares its observed distribution with specified benchmark probabilities.

Example:

Does customer channel mix match 50% store, 30% website, and 20% app?

The distinction is:

Method	Variables	Question
Chi-square independence	2 categorical variables	Are the variables associated?
Chi-square goodness-of-fit	1 categorical variable	Does its distribution match specified proportions?

82 Expected Counts for Goodness-of-Fit

Suppose:

$$n = 500$$

and the benchmark probabilities are:

$$p_{Store} = 0.50$$

$$p_{Website} = 0.30$$

$$p_{App} = 0.20$$

Expected counts are:

$$E_i = np_{i,0}$$

Therefore:

82.0.1 Store

$$E_{Store} = 500(0.50) = 250$$

82.0.2 Website

$$E_{Website} = 500(0.30) = 150$$

82.0.3 App

$$E_{App} = 500(0.20) = 100$$

So under the benchmark:

Channel	Observed	Expected
Store	240	250
Website	160	150
App	100	100

The chi-square statistic again measures how far observed counts are from expected counts.

83 Goodness-of-Fit Chi-Square Statistic

The statistic is:

$$X^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

where:

$$E_i = np_{i,0}$$

The logic is identical to the independence test:

Compare what we observed with what we would expect if the null model were correct.

84 Goodness-of-Fit Hypotheses

For the channel example:

$$H_0 : \begin{cases} p_{Store} = 0.50 \\ p_{Website} = 0.30 \\ p_{App} = 0.20 \end{cases}$$

The alternative is:

H_a : The population distribution differs from the benchmark in at least one category

85 Goodness-of-Fit Degrees of Freedom

If there are:

$$k$$

categories and their probabilities are specified in advance, then:

$$df = k - 1$$

For three channels:

$$df = 3 - 1 = 2$$

86 Goodness-of-Fit in R

Suppose:

```
channel_counts <- c(  
  Store = 240,  
  Website = 160,  
  App = 100  
)
```

The test can be performed with:

```
chisq.test(  
  channel_counts,  
  p = c(0.50, 0.30, 0.20)  
)
```

For the example:

$$X^2 = 1.0667$$

$$df = 2$$

and:

$$p\text{-value} = 0.5866$$

Because the p-value is not small, we fail to reject the null hypothesis.

The data do not provide sufficient evidence that the population channel distribution differs from:

$$50\%, 30\%, 20\%$$

This does **not** prove that the true population proportions are exactly:

50%, 30%, 20%

It means the observed differences are not sufficiently inconsistent with the benchmark to reject it.

87 One Logic Behind Both Chi-Square Tests

Both chi-square procedures follow the same basic structure:

$$\boxed{\text{Observed} \quad \text{vs.} \quad \text{Expected under } H_0}$$

and both use:

$$\boxed{X^2 = \sum \frac{(O - E)^2}{E}}$$

What changes is **where the expected counts come from.**

87.0.1 Test of Independence

Expected counts are determined from the row and column totals:

$$E_{ij} = \frac{(\text{Row Total}_i)(\text{Column Total}_j)}{\text{Grand Total}}$$

87.0.2 Goodness-of-Fit

Expected counts come from specified benchmark probabilities:

$$E_i = np_{i,0}$$

So the mathematical logic is the same, but the null model is different.

88 Choosing the Appropriate Categorical Method

The first step is identifying the business question.

Business Question	Analysis
How much do two conversion rates differ?	Two-proportion CI and test
Does conversion differ across three pages?	Chi-square test on a two-way table
Are homeownership and marital status associated?	Chi-square test of independence
Does channel mix match specified shares?	Chi-square goodness-of-fit test

A useful decision process is:

88.0.1 Are You Comparing Two Population Proportions?

For example:

Does the new page have a different conversion rate from the current page?

Use:

Two-Proportion Inference

88.0.2 Are You Examining Two Categorical Variables?

For example:

Is purchase outcome associated with which page a customer sees?

or:

Is homeownership associated with marital status?

Use:

Chi-Square Test of Independence

88.0.3 Are You Comparing One Categorical Variable to Predetermined Proportions?

For example:

Does the observed channel mix match the expected 50/30/20 split?

Use:

Chi-Square Goodness-of-Fit Test

89 Categorical Inference Workflow

When working with categorical data:

89.1 Step 1: Identify the Business Question

Determine what comparison is actually being requested.

89.2 Step 2: Identify the Observational Unit

Ask:

What does one observation represent?

This is essential for determining whether observations are independent.

89.3 Step 3: Identify the Categorical Variables

Determine whether the problem contains:

- One categorical variable
 - Two categorical variables
 - Two groups with a binary outcome
-

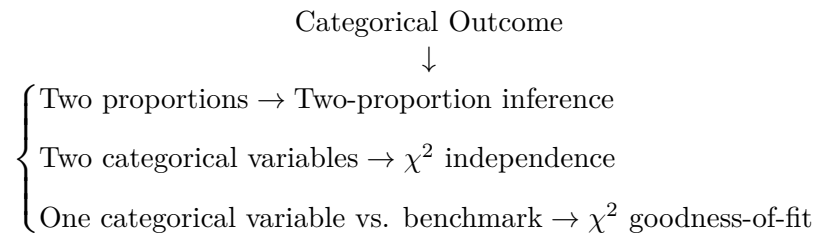
89.4 Step 4: Examine Counts and Proportions

Construct:

- Frequency tables
- Contingency tables
- Conditional proportions
- Bar charts when useful

Do not interpret raw counts without considering group sizes.

89.5 Step 5: Choose the Appropriate Method



89.6 Step 6: State the Null Model

For two proportions:

$$H_0 : p_1 = p_2$$

For independence:

$$H_0 : \text{The categorical variables are independent}$$

For goodness-of-fit:

$$H_0 : \text{The category probabilities equal the specified benchmark}$$

89.7 Step 7: Check Conditions

Depending on the method, examine:

- Independence of observations
 - Mutually exclusive categories
 - Expected counts
 - Sample sizes
-

89.8 Step 8: Conduct the Analysis

Use the appropriate R function:

```
prop.test()
```

or:

```
chisq.test()
```

89.9 Step 9: Interpret in Context

Translate the statistical result back into the business question.

Avoid saying only:

The p-value is significant.

Instead explain **what statistical relationship the data support**.

89.10 Step 10: Separate Association from Causation

A statistically significant association does not automatically establish causality.

Ask:

How were the data generated?

Random assignment can support causal interpretation.

Observational association by itself cannot.

90 Connecting Lectures 5–8

The inferential methods covered so far can now be organized according to the **type of outcome and study design**.

Question	Parameter / Structure	Method
What is one population mean?	μ	One-sample t
What is the mean difference for linked observations?	μ_d	Paired t
How do two independent population means differ?	$\mu_1 - \mu_2$	Welch two-sample t
How do two population proportions differ?	$p_1 - p_2$	Two-proportion inference
Are two categorical variables associated?	Contingency table	Chi-square independence
Does one categorical distribution match specified shares?	$p_1, \dots, p_k$	Chi-square goodness-of-fit

The broader decision process is:

Business Question → Outcome Type → Study Design → Parameter → Statistical Method → Interpretation

The key idea remains the same:

Do not start by choosing a statistical test. Start by understanding what the variables represent, how the data were generated, and what population question you are trying to answer.